LUMORA: COGNITIVE-STYLE AND COGNITIVE-LOAD-AWARE ADAPTIVE LEARNING

INDIVIDUAL THESIS

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Individual component: Learner Profile and Cognitive Style Detection

Group ID: R26-SE-004

Report submitted in partial fulfilment of the requirements for the

Bachelor of Science Honours in Information Technology

Specialisation in Software Engineering

Department of Computer Science and Software Engineering

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

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ABSTRACT

Understanding learners' cognitive preferences can support the development of more personalized and adaptive online learning environments. However, identifying visual–verbal cognitive style through questionnaires alone may require additional effort from learners. This research presents a behavioral-based Visual–Verbal Cognitive Style Prediction System designed to predict questionnaire-derived cognitive style categories using learners' interactions with visual and verbal learning materials. The proposed system integrates a web-based learning interface, webcam-based gaze estimation, cursor and scrolling activity collection, behavioral feature extraction, and supervised machine learning.

The learning interface was developed using React and Vite and presents image-based and text-based content in separate panels with independent scrolling. Behavioral information, including image-associated cursor activity, scrolling activity, gaze-related behavior, and first interaction preference, is collected during the learning activity. MediaPipe FaceLandmarker is used to estimate gaze-related information from facial and iris landmarks. Following initial observations of inconsistent gaze estimates, a personalized multi-stage calibration process was introduced to check webcam positioning and establish individualized gaze-coordinate ranges. The collected behavioral data are processed and used to train a Support Vector Machine classifier with an RBF kernel. Data preprocessing includes missing-value filtering, categorical encoding, feature standardization, stratified train–test splitting, and training-only SMOTE to address class imbalance.

The system was developed and deployed for research data collection, with undergraduate participants from the Faculty of Science and the Faculty of Management and IT at the University of Kelaniya. Model performance is evaluated using accuracy, precision, recall, F1-score, weighted F1-score, and a confusion matrix. The findings and numerical evaluation results are presented based on the actual experimental outputs. The proposed system demonstrates an approach to investigating the relationship between observable learner behavior and questionnaire-derived visual–verbal cognitive style categories.

However, webcam-based gaze estimates remain approximate, and the effectiveness of personalized calibration requires separate empirical validation. The predictions should therefore be interpreted as model-generated estimates rather than definitive measurements of learners' cognitive characteristics. Future work includes improving gaze-estimation reliability, expanding the participant dataset, comparing alternative machine learning models, and validating the system across different learning environments.

Keywords: Visual–verbal cognitive style, behavioral analysis, gaze estimation, personalized calibration, Support Vector Machine, adaptive learning.

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my research supervisor, Dr Kalpani Manathunga, and co-supervisor, Mr Jeewaka Perera, for their guidance, supervision, and valuable feedback throughout this research project.

I would also like to acknowledge my team members, Dilshan W.G. W. and Subasinghe M.H.R.I., for their collaboration and contributions to the wider research project. Their work on explainable artificial intelligence, cognitive style detection, and cognitive load detection provides the broader context for the development of the proposed learning system.

I extend my appreciation to the University of Kelaniya and the undergraduate participants who contributed to the research data collection. Their participation supported the investigation of behavioral patterns associated with visual–verbal cognitive style.

Finally, I acknowledge the Sri Lanka Institute of Information Technology (SLIIT) for providing the academic environment in which this research was conducted and submitted. I am grateful to everyone who supported the successful completion of this research.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence [1]
API	Application Programming Interface [1]
ASR	Automatic Speech Recognition [1]
FIFO	First-In, First-Out [1]
FRE	Flesch Reading Ease [1]
HF	Hugging Face [1]
HTTP	Hypertext Transfer Protocol [1]
JWT	JSON Web Token [1]
LLM	Large Language Model [1]
MCQ	Multiple-Choice Question [1]
OMML	Office Math Markup Language [1]
PDF	Portable Document Format [2]
PPTX	Office Open XML Presentation Format [2]
RAG	Retrieval-Augmented Generation [2]
SBERT	Sentence-BERT [2]
SDG	Sustainable Development Goal [2]
TF	Term Frequency [2]
XML	Extensible Markup Language

1 INTRODUCTION

1.1 Overall project context

Lumora is a multi-component adaptive learning platform intended to connect educational content with a learner's cognitive characteristics and changing learning effort. Teachers create courses and upload lecture materials. Students interact with video lessons and receive generated learning aids. Cognitive style detection uses webcam and cursor behavioral information to estimate presentation-related and analytic–wholistic characteristics. Cognitive load detection uses video interaction features, including pausing and rewinding, to estimate learning effort. Explainable artificial intelligence (AI) presents model-related insights to teachers, while course-linked exam aids provide practice questions from educator resources.

The full platform therefore requires more than prediction models. It needs a content infrastructure that can turn a teacher's heterogeneous files into usable evidence and then pass that evidence to the adaptation process. Without this infrastructure, a model may generate a readable explanation that does not represent the lecture, or it may receive duplicated and damaged material that compromises the adaptation. The individual component addressed in this thesis supplies this resource-to-explanation connection.

The complete application uses a React interface, an Express application programming interface (API) gateway, and separate Node.js and Python services. Course and resource information are represented in MongoDB. Other components use their own services and databases. This thesis discusses those neighboring components only where they provide inputs or receive outputs; it does not claim ownership of their cognitive prediction or explanation algorithms.

1.2 Individual component context

Learner profile detection and cognitive style analysis play an important role in adaptive learning systems by identifying individual learners' learning approaches and cognitive characteristics. Understanding these characteristics can help an adaptive learning platform select and present educational content according to learners' individual needs and preferences.

This research component focuses on learner profile detection through three main areas: learning approaches, Visual–Verbal cognitive styles, and Analytic–Wholistic cognitive styles. Learning approaches are identified using the ASSIST questionnaire, which classifies learners into Surface, Deep, and Strategic approaches to learning. Similarly, Analytic–Wholistic cognitive styles are assessed using a questionnaire designed to identify whether learners tend towards analytic or wholistic information processing.

In contrast, Visual–Verbal cognitive style detection is investigated using a supervised machine learning model trained on behavioral data associated with reference labels obtained from a standardized cognitive-style questionnaire. This approach aims to predict learners' Visual–Verbal cognitive styles from their behavioral patterns, reducing the need to administer the reference questionnaire for every prediction.

The component combines questionnaire-based assessments and machine learning techniques to develop a comprehensive learner profile. The profile incorporates learners' ASSIST learning approach classifications, questionnaire-based Analytic–Wholistic cognitive style assessments, and machine learning-based Visual–Verbal cognitive style predictions.

The resulting learner profile is intended to support the wider adaptive learning platform in personalizing educational content and learning experiences. The accuracy and reliability of the classification methods, as well as the usefulness of the resulting profiles for supporting personalized learning, will be evaluated through appropriate validation and testing.

1.3 Problem statement

Learners differ in their approaches to studying and in how they process and engage with educational information. If an educational platform provides the same learning materials and presentation methods to every learner, it may not adequately account for these differences. Developing a learner profile can help the platform consider relevant individual characteristics when supporting learning activities.

However, identifying these characteristics presents several challenges. Questionnaire-based assessments require learners to spend time answering questions, and longer assessments may introduce additional effort into the learning process. In addition, relying

entirely on self-reported responses may limit the availability of profile information when repeated assessment is impractical.

Machine learning offers a potential approach for identifying Visual–Verbal cognitive-style patterns from behavioral data. Nevertheless, such a model requires suitable labeled training data and systematic evaluation to determine whether its predictions are sufficiently reliable. Similarly, questionnaire-based classification of learning approaches and Analytic–Wholistic cognitive styles requires appropriate scoring procedures and careful interpretation of the resulting categories.

Therefore, the problem addressed by this research component is how to combine questionnaire-based classification and behavior-based machine learning to develop a learner profile that identifies selected learning approaches and cognitive styles while considering assessment effort, classification reliability, and potential use in adaptive educational content delivery.

1.4 Research gap

Existing research has explored learner profiling, cognitive style detection, and adaptive learning systems using questionnaire-based assessments, behavioral data, and advanced sensing technologies. These approaches provide a foundation for understanding individual learner characteristics and adapting educational content accordingly. However, several research gaps remain in developing practical and integrated learner-adaptive systems.

First, some existing approaches to cognitive style detection rely on specialized technologies, such as electroencephalography (EEG) and advanced eye-tracking equipment, to collect information about learners' cognitive processes and visual attention. Although these technologies can provide valuable data, their hardware requirements, implementation costs, and practical constraints may limit their use in ordinary educational environments. Therefore, there is a need to investigate more accessible approaches that use commonly available devices, such as webcams, together with mouse cursor behavioral data to predict cognitive styles without requiring specialized sensing hardware.

Second, behavioral data collected through webcams and cursor interactions may provide useful information about learners' interaction patterns during digital learning activities.

However, identifying cognitive styles from these signals remains a challenge because observable behaviors do not directly establish a learner's cognitive characteristics. Appropriate reference labels, supervised machine learning techniques, and systematic model evaluation are therefore necessary to investigate the reliability of such predictions.

Third, learner profiling and cognitive style detection are often addressed as separate aspects of personalized learning. A learner profile may describe an individual's learning approach, while cognitive style analysis focuses on how the learner tends to process and represent information. When these characteristics are considered independently, the adaptive learning system may have limited information for making personalization decisions. An integrated approach that considers both learning approaches and cognitive styles could provide a more comprehensive basis for selecting and presenting educational materials.

To address these gaps, this research develops and evaluates an integrated learner profiling and cognitive style analysis component. The ASSIST questionnaire is used to classify learners into Surface, Deep, and Strategic learning approaches, while a questionnaire-based assessment is used to identify Analytic–Wholistic cognitive styles. In addition, a supervised machine learning model investigates the prediction of Visual–Verbal cognitive styles using behavioral data, including webcam-derived features and mouse cursor interactions, with reference labels obtained from a standardized cognitive-style questionnaire.

The main contribution of this research is the implementation and evaluation of a combined approach that incorporates questionnaire-based learner profiling, behavioral-data-based cognitive style prediction, and questionnaire-based cognitive style assessment. By bringing these characteristics together in a unified learner profile, the component aims to provide the wider adaptive learning platform with information that can support more individualized educational content delivery. The reliability of the predictions and the practical usefulness of the integrated profile will be assessed through appropriate evaluation rather than assumed in advance.

1.5 Aim and objectives

Aim

To develop and evaluate an integrated learner profiling and cognitive style analysis component that combines questionnaire-based assessments and supervised machine learning using webcam-derived and mouse cursor behavioral data to identify selected learning approaches and cognitive styles, supporting personalized educational content delivery.

Objectives

1. To implement questionnaire-based learner profiling using the ASSIST questionnaire to classify learners into Surface, Deep, and Strategic learning approaches according to the appropriate scoring procedure.
2. To develop and evaluate a supervised machine learning model for predicting Visual–Verbal cognitive styles using webcam-derived features and mouse cursor behavioral data, with reference labels obtained from a standardized cognitive-style questionnaire.
3. To implement questionnaire-based assessment of Analytic–Wholistic cognitive styles using an appropriate assessment instrument and scoring procedure.
4. To integrate the identified learning approaches and cognitive style classifications into a unified learner profile that can support personalized educational content selection and presentation within the wider adaptive learning platform.
5. To evaluate the reliability and effectiveness of the implemented component by assessing machine learning prediction performance, questionnaire scoring correctness, and overall system functionality.
6. To investigate the practicality of using accessible behavioral data, including webcam and mouse cursor interactions, for cognitive style prediction without requiring specialized sensing hardware such as dedicated eye-tracking equipment or EEG devices.
7. To assess the potential usefulness of combining learner profiles and cognitive style classifications as a basis for supporting adaptive educational content delivery.

1.6 Research questions or hypotheses

This research is guided by the following questions:

RQ1: How can the ASSIST questionnaire be used to classify learners into Surface, Deep, and Strategic learning approaches?

RQ2: To what extent can webcam-derived features and mouse cursor behavioral data be used to predict Visual–Verbal cognitive styles using supervised machine learning?

RQ3: How can questionnaire-based assessment be used to classify learners into Analytic–Wholistic cognitive styles?

RQ4: How can learner profiles and cognitive style classifications be integrated to support personalized educational content delivery?

RQ5: How effectively does the proposed component perform in terms of classification accuracy, prediction performance, and system functionality?

These questions will be addressed through implementation and evaluation of the proposed component.

1.7 Scope, limitations and assumptions

Scope

This research focuses on developing a learner profiling and cognitive style analysis component for a wider adaptive learning platform. Its scope includes:

- Classification of Surface, Deep, and Strategic learning approaches using the ASSIST questionnaire.
- Prediction of Visual–Verbal cognitive styles using supervised machine learning based on webcam-derived features and mouse cursor behavioral data.
- Classification of Analytic–Wholistic cognitive styles using a questionnaire-based assessment.
- Integration of the identified learning approaches and cognitive styles into a unified learner profile to support personalized educational content delivery.
- Evaluation of classification performance, questionnaire scoring, and system functionality.

Limitations

Questionnaire-based classifications depend on the suitability of the assessment instruments and the accuracy of learner responses. Machine learning performance depends on the quality, quantity, and representativeness of the available behavioral data and reference labels.

Webcam and mouse cursor behaviors may be influenced by factors other than cognitive styles, such as device differences, task difficulty, and individual interaction habits. Therefore, predictions may not always accurately represent learners' cognitive styles. The model's generalizability may also be limited by the available training data.

Furthermore, developing learner profiles does not automatically guarantee improved learning outcomes. The effectiveness of using these profiles for educational personalization requires further evaluation.

Assumptions

This research assumes that the selected questionnaires and their scoring procedures are appropriate for identifying the intended learning approaches and cognitive styles. It also assumes that questionnaire-derived reference labels are suitable for training the Visual–Verbal prediction model.

The research further assumes that webcam-derived features and mouse cursor behavioral data may contain useful information for predicting Visual–Verbal cognitive styles, which will be investigated through model evaluation. Finally, it assumes that the resulting learner profile can be integrated into the wider adaptive learning platform to support personalized educational content delivery.

1.8 Individual contribution

The individual contribution of this research is the design, implementation, and evaluation of a learner-profiling component that combines questionnaire-based classification with behavior-based machine learning.

The component includes the implementation of ASSIST-based scoring to classify learning approaches, the development of a supervised machine learning approach for Visual–Verbal cognitive-style prediction using behavioral data and reference labels, and the implementation of questionnaire-based classification for Analytic–Wholistic cognitive styles.

The individual work also includes integrating the resulting classifications into a structured learner profile and evaluating the relevant scoring procedures, model performance, and software functionality. This contribution provides the wider adaptive learning platform with learner-profile information that may be used to inform educational content personalization.

The contribution is limited to the work implemented and evaluated within this individual component. Any claims concerning improvements in learning outcomes or the overall effectiveness of the complete platform require separate supporting evidence.

1.9 Thesis structure

This thesis is organized into chapters that describe the background, design, implementation, and evaluation of the proposed learner-profiling component.

Chapter 1 – Introduction: Presents the research context, problem statement, research gap, aim and objectives, research questions, scope, limitations, assumptions, and individual contribution.

Chapter 2 – Literature Review: Reviews existing research on adaptive learning, learner modeling, learning approaches, Visual–Verbal cognitive styles, Analytic–Wholistic cognitive styles, questionnaire-based assessment, and machine learning approaches for identifying learner characteristics.

Chapter 3 – Methodology: Describes the research design, selected assessment instruments, questionnaire scoring procedures, behavioral data collection, machine learning development process, system architecture, and evaluation methods.

Chapter 4 – Implementation: Explains the implementation of the questionnaire-based classification methods, the Visual–Verbal prediction model, and the integration of classification results into the learner profile.

Chapter 5 – Results and Evaluation: Presents the results of the classification and software evaluations, including relevant machine learning performance measures, scoring tests, and functional testing.

Chapter 6 – Discussion and Conclusion: Interprets the findings in relation to the research questions, discusses the contribution and limitations of the component, and identifies possible directions for future work.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This literature review examines research related to learner profiling, learning approaches, cognitive styles, questionnaire-based assessment, and machine learning techniques for identifying learner characteristics in digital learning environments. The purpose of the review is to establish the theoretical and technical foundation for developing a learner-profiling component that can support personalized educational content delivery.

The review focuses on three learner characteristics investigated in this research. The first is learning approach, classified into Surface, Deep, and Strategic categories using the ASSIST questionnaire. The second is Visual–Verbal cognitive style, for which a supervised machine learning model is developed using behavioral data and reference labels obtained from a standardized cognitive-style questionnaire. The third is Analytic–Wholistic cognitive style, assessed through a questionnaire-based classification method.

Relevant literature can be organized into four main areas: theoretical studies of learning approaches and cognitive styles, studies of questionnaire-based learner assessment, research on the implicit identification of learner characteristics from behavioral data, and existing adaptive learning systems that use learner models to inform educational content delivery.

The review considers the purpose and suitability of different assessment methods, the role of behavioral features in machine learning, and the challenges associated with classification reliability, assessment effort, usability, and generalizability. It also considers research questioning whether matching educational instruction to a preferred learning style consistently improves learning outcomes. This distinction is important because identifying a learner characteristic and demonstrating the educational benefit of adapting content to that characteristic are separate research objectives.

For academic reliability, the literature review should prioritize peer-reviewed journal articles, conference papers, established theoretical publications, and original documentation for relevant assessment instruments and systems. The final submitted version should document the databases searched, search terms, publication-date range,

inclusion and exclusion criteria, and screening process actually used. These details should be reported from the review process conducted by the researcher rather than assumed or reconstructed.

2.2 Domain and theoretical foundations

2.2.1 Adaptive Learning and Learner Modeling

Adaptive learning systems aim to modify aspects of an educational experience according to information about the learner, the learning task, or the learner's progress. A learner model represents selected characteristics that can help an educational system make decisions about presenting material, recommending activities, or providing support.

Chrysafiadi and Virvou (2013) reviewed student modeling approaches and discussed the importance of representing learner information within adaptive educational systems. A learner model may include knowledge, goals, preferences, learning behavior, and other relevant characteristics, depending on the purpose of the system.

For the present research, the learner profile represents three selected characteristics: learning approach, Visual–Verbal cognitive style, and Analytic–Wholistic cognitive style. These characteristics are intended to provide information that the wider platform can use when considering how educational content should be presented. They should be treated as inputs to personalization rather than complete descriptions of a learner.

2.2.2 Learning Approaches: Surface, Deep, and Strategic

Learning approaches describe how students engage with learning tasks and organize their study activities. The Surface, Deep, and Strategic categories provide a framework for understanding different approaches to academic learning.

A Surface approach generally emphasizes meeting immediate task requirements, memorization, and limited engagement with underlying meaning. A Deep approach emphasizes understanding concepts, relating ideas, and seeking meaning. A Strategic approach emphasizes organizing study activities and managing effort to achieve academic goals. These categories describe approaches to learning rather than fixed types of people; the same learner may use different approaches in different circumstances.

The ASSIST questionnaire is associated with research on approaches and study skills in higher education. Tait, Entwistle, and McCune (1998) reported work on the development and use of the revised Approaches and Study Skills Inventory for Students. The instrument provides a structured method for assessing learning-related approaches through questionnaire responses.

In this research, ASSIST is used to classify learners into Surface, Deep, and Strategic categories. The resulting classification provides one dimension of the learner profile. The validity of the classification depends on using the appropriate questionnaire version, following its scoring guidance, and interpreting the results within the instrument's intended context.

2.2.3 Visual–Verbal Cognitive Styles

Visual–Verbal cognitive style concerns differences in how people represent or process information, particularly through visual representations and verbal or linguistic information. Visual representations may include diagrams, illustrations, and spatial arrangements, whereas verbal representations may include written explanations and text-based descriptions.

Mayer and Massa (2003) examined visual and verbal learners in relation to different facets of learning and cognitive processing. Their work illustrates that the relationship between visual and verbal processing is more complex than simply assigning each learner to a fixed category.

In the proposed component, Visual–Verbal cognitive style is approached as a classification task. A standardized cognitive-style questionnaire provides reference labels for the behavioral data used to train a supervised machine learning model. Once trained and evaluated, the model is intended to predict the relevant category from behavioral patterns without requiring the same reference questionnaire to be completed for every prediction.

This approach introduces an important methodological distinction: the questionnaire-derived label is the target used to train and evaluate the model, while the behavioral data are the input features. The model's performance therefore indicates how well it predicts the

questionnaire-derived classification, not necessarily whether it measures every aspect of visual or verbal information processing.

2.2.4 Analytic–Wholistic Cognitive Styles

Analytic–Wholistic cognitive style concerns differences in how information and relationships are processed. Analytic processing emphasizes individual elements and their attributes, whereas wholistic processing emphasizes relationships, context, and the broader configuration of information.

Research by Choi, Koo, and Choi (2007) investigated analytic–holistic thinking and its measurement. Their work provides a foundation for examining differences in how individuals attend to objects, contexts, and relationships. However, analytic–holistic thinking should not automatically be treated as interchangeable with other cognitive-style dimensions, as different instruments may measure different constructs.

This research includes an Analytic–Wholistic classification as another dimension of the learner profile. The classification is obtained through a questionnaire-based assessment and its associated scoring procedure. The selected instrument, scoring rules, and interpretation of its categories should be specified clearly in the methodology chapter.

2.2.5 Behavioral Data and Supervised Machine Learning

Digital learning environments can record interactions such as mouse movement, clicking, navigation, time spent on tasks, and responses to educational activities. Depending on the learning task and data-collection design, these interactions may provide information relevant to a learner's engagement or task behavior.

Machine learning techniques can be used to identify patterns in such data. In supervised learning, a model is trained using input examples paired with known target labels. For this research, the behavioral data form the model inputs, and the reference cognitive-style questionnaire supplies the target labels for the Visual–Verbal classification task.

A reliable evaluation requires separating training data from data used for testing. Where appropriate, a validation set or cross-validation can support model selection. Performance should be examined using suitable measures, including precision, recall, F1-score, and a

confusion matrix, rather than relying on accuracy alone. The selection of metrics should account for class balance and the practical consequences of different classification errors.

The model also needs to be assessed for overfitting and generalizability. If observations from the same participant appear in both training and testing sets, the evaluation may overestimate performance when the intended use is prediction for previously unseen learners. Where multiple observations belong to one learner, a participant-level split may therefore be more appropriate.

2.3 Related research

2.3.1 Questionnaire-Based Learner Profiling

Questionnaires are commonly used to assess learner characteristics because they provide structured responses that can be scored according to an established procedure. The ASSIST framework, for example, provides a means of examining students' approaches to learning.

Questionnaire-based methods offer a direct and interpretable route to classification. However, their usefulness depends on the relevance of the instrument, the quality of the responses, and the appropriateness of its scoring procedure. Completing multiple assessments may also increase the effort required from learners.

For this research, questionnaires serve two roles. ASSIST provides the basis for learning-approach classification, while standardized cognitive-style questionnaires provide reference labels for Visual–Verbal model development and responses for Analytic–Wholistic classification. These uses should be distinguished clearly because a questionnaire used to generate model-training labels does not necessarily need to be used during every future model prediction.

2.3.2 Research on Implicit Learner Characteristic Detection

Implicit learner modeling investigates whether learner characteristics can be inferred from data generated during interactions with a digital learning environment. Lwande, Muchemi, and Oboko (2021) reviewed research on identifying learner styles and cognitive traits implicitly in e-learning platforms, providing relevant context for approaches that use learner interaction data rather than relying exclusively on direct self-report.

Behavior-based approaches may reduce the need to repeatedly ask learners to complete a particular questionnaire. Nevertheless, the presence of behavioral data does not guarantee that a cognitive characteristic can be inferred accurately. The observed behavior may be affected by task difficulty, familiarity with the interface, accessibility requirements, device characteristics, and other situational factors.

The proposed research investigates this issue through supervised learning for Visual–Verbal classification. Its contribution must therefore be assessed empirically by comparing model predictions with suitable reference labels on data not used for model training.

2.3.3 Research on Visual and Verbal Processing

Research on visual and verbal processing has explored individual differences in the representation and comprehension of information. Mayer and Massa (2003) examined multiple facets of visual and verbal learners, while other studies have investigated how learners interact with different representations.

Such research provides theoretical context for examining visual and verbal processing, but it does not automatically establish that each learner benefits most from receiving material that matches a presumed style. A classification model may identify patterns associated with a questionnaire-defined category without demonstrating that matching content to that category improves learning.

Pashler et al. (2008) reviewed the evidence for learning-style concepts and highlighted the need for rigorous experimental evidence before claiming that matching instruction to a learner's style improves outcomes. Kirschner and van Merriënboer (2013) also discussed misconceptions in education, including assumptions about learners knowing which instructional methods are best for them.

Accordingly, the present research treats Visual–Verbal classification as a learner-profiling task. Any claim that its use improves learning performance would require further evaluation involving appropriate instructional comparisons and learning-outcome measures.

2.3.4 Research on Analytic and Wholistic Processing

Research on analytic and wholistic thinking has examined differences in attention to focal objects, background context, and relationships among elements. Choi, Koo, and Choi (2007) investigated the measurement of analytic–holistic thinking, offering relevant background for questionnaire-based assessment of this dimension.

The choice of assessment instrument matters because different tools may operationalize analytic and wholistic processing differently. A classification produced by one instrument should therefore be interpreted according to that instrument's definitions and scoring procedure.

In this project, the Analytic–Wholistic dimension complements the other profile dimensions. Its role is to contribute additional learner information to the overall profile, rather than to serve as a substitute for learning-approach classification or Visual–Verbal prediction.

2.3.5 Synthesis of Related Research

The reviewed areas suggest that questionnaire-based assessment and behavior-based machine learning can serve complementary roles in learner profiling. Questionnaires provide structured assessments and reference labels, while behavioral data offer a possible means of making predictions without repeating a specific questionnaire at every interaction.

However, these approaches have different sources of uncertainty. Questionnaire classifications depend on the instrument and scoring procedure. Machine learning predictions depend on the training data, label quality, selected features, model design, and evaluation methodology. Combining several classifications also requires careful interpretation so that the learner profile does not imply greater certainty than the individual methods support.

The present research brings these approaches together in one component, with the aim of producing a structured profile that can support personalization within the wider learning platform.

2.4 Existing systems, products and techniques

2.4.1 Adaptive Learning Systems

Adaptive learning systems use learner-related information to make decisions about learning activities, resources, or instructional support. Student modeling research, such as the review by Chrysafiadi and Virvou (2013), provides a basis for understanding how learner information can be represented and used in adaptive educational systems.

Systems differ in the information they collect and the decisions they make from it. Some focus primarily on learner knowledge and progress, while others incorporate preferences, learning behavior, or additional learner characteristics. The inclusion of a particular characteristic in a learner model does not, on its own, establish that it is useful for improving learning outcomes.

For the proposed platform, the learner profile is intended to supply selected classifications that can inform the presentation of educational content. The extent to which those classifications change content selection or presentation depends on the integration implemented in the wider system.

2.4.2 Questionnaire-Based Assessment Techniques

Questionnaire-based assessment provides a structured way to collect information about learner characteristics. The ASSIST questionnaire is relevant to the classification of Surface, Deep, and Strategic learning approaches. Standardized cognitive-style questionnaires can also provide reference labels for machine learning and direct assessment results for other profile dimensions.

The main strengths of questionnaire-based techniques are their structured response formats and the possibility of applying documented scoring procedures. Their limitations include completion effort, dependence on self-report, and the need to interpret results according to the intended construct.

The proposed component uses questionnaire-based techniques where they are required for learning-approach classification, reference-label generation, and Analytic–Wholistic assessment. The reference questionnaire used to label training data has a different

operational role from the model used to predict Visual–Verbal categories from behavioral inputs.

2.4.3 Behavioral Feature-Based Classification

Behavioral feature-based classification uses measurable interaction data as input to a predictive model. Potential feature categories in digital learning environments include interaction frequency, navigation behavior, task timing, and mouse or cursor movement. The relevance of any specific feature depends on the tasks, instrumentation, and data actually available in the proposed system.

A typical supervised classification pipeline includes data collection, preprocessing, feature construction, label assignment, model training, validation, and testing. The evaluation stage examines whether the model generalizes to data outside the training set.

For this research, the behavioral features must be defined in the methodology chapter according to the variables collected by the implemented system. The model's performance should then be reported using results from the actual experiments. Specific feature importance claims, accuracy figures, or comparisons with other algorithms should not be included unless they are supported by experimental results or verified literature.

2.4.4 Assessment Usability and Practical Considerations

The practical suitability of a learner-profiling technique depends not only on classification performance but also on its completion effort, accessibility, and integration into the learning environment. Longer assessments may impose additional effort, while short assessments may not capture a construct as comprehensively as a longer instrument. Therefore, reducing questionnaire length should not be treated as an improvement unless the alternative assessment is suitable for the intended purpose.

The selection of an Analytic–Wholistic assessment should consider the instrument's construct, scoring procedure, evidence of reliability and validity, completion time, and suitability for the platform's users and devices. These criteria are particularly relevant if the research seeks to reduce assessment effort while maintaining meaningful classification.

Similarly, behavioral prediction may reduce the need for repeated questionnaire completion for Visual–Verbal classification, but it introduces requirements for data quality, privacy-conscious collection, and model validation. It should be considered a complementary technique rather than an automatic replacement for standardized assessment in every context.

2.4.5 Comparison of Relevant Techniques

The techniques considered in this research serve different purposes:

Technique	Main purpose	Main consideration
ASSIST questionnaire	Classifies Surface, Deep, and Strategic learning approaches	Appropriate questionnaire version and scoring procedure
Standardized Visual–Verbal questionnaire	Provides reference labels for model training and evaluation	Suitability of labels for the intended classification task
Supervised machine learning	Predicts Visual–Verbal category from behavioral features	Generalization, class balance, and evaluation performance
Analytic–Wholistic questionnaire	Classifies the Analytic–Wholistic dimension	Instrument validity, scoring, and assessment effort
Integrated learner profile	Combines the three classification results	Clear interpretation and appropriate use in personalization

Table 1 Different Techniques used in the Research

This comparison shows that the component does not rely on one universal technique for all learner characteristics. Instead, it uses methods suited to the intended role of each classification process.

2.5 Gap synthesis and conceptual framework

2.5.1 Synthesis of the Research Gap

The literature provides foundations for questionnaire-based learner assessment, cognitive-style research, supervised machine learning, and learner modeling in adaptive learning

environments. Nevertheless, the proposed component addresses a practical integration problem involving three specific learner characteristics.

First, learning approaches and cognitive styles represent different constructs and should not be treated as interchangeable. A learner profile that includes all three must preserve the meaning of each dimension and use appropriate assessment or prediction methods for each one.

Second, questionnaire-based assessment provides structured classifications but may require learners to complete additional questions. Behavioral machine learning offers a possible alternative for predicting Visual–Verbal categories after a model has been trained, but the quality of those predictions must be demonstrated using suitable evaluation data.

Third, the combination of multiple learner characteristics raises questions about the reliability and practical use of the resulting profile. Each classification process requires its own appropriate evaluation, and the overall profile should not be considered valid merely because its components have been combined.

Finally, producing a learner profile is not equivalent to demonstrating the effectiveness of personalized instruction. Evidence that the profile accurately represents the intended classifications must be distinguished from evidence that using it improves learning outcomes.

The research therefore focuses on implementing and evaluating an integrated learner-profiling component that combines ASSIST-based learning-approach classification, behavior-based Visual–Verbal cognitive-style prediction, and questionnaire-based Analytic–Wholistic classification.

2.5.2 Proposed Conceptual Framework

The conceptual framework consists of three classification processes that contribute to a combined learner profile.

Process 1: Learning-approach classification. Learner responses to the ASSIST questionnaire are processed using the relevant scoring procedure to obtain a Surface, Deep, or Strategic classification.

Process 2: Visual–Verbal prediction. Behavioral data are processed into model inputs. A supervised machine learning model trained using questionnaire-derived reference labels predicts the Visual–Verbal category. The model is evaluated against appropriate reference labels using a test procedure designed to assess generalization.

Process 3: Analytic–Wholistic classification. Responses to the selected questionnaire are processed using the instrument's scoring rules to obtain an Analytic–Wholistic classification.

The outputs of these three processes are combined into a structured learner profile. The wider adaptive learning platform may then use this profile as one source of information when selecting or presenting educational materials. The actual educational benefit of that use remains a question for evaluation.

2.5.3 Relationship Between the Framework and the Research Objectives

The framework maps directly to the objectives of the research. The ASSIST process supports the objective of classifying learning approaches. The behavioral machine learning process supports the objective of predicting Visual–Verbal cognitive style. The questionnaire-scoring process supports the objective of classifying Analytic–Wholistic cognitive style. Combining these outputs supports the objective of constructing a learner profile for the wider platform.

The framework also identifies separate evaluation requirements. Questionnaire-based scoring should be checked against the relevant scoring rules, the machine learning model should be evaluated using suitable classification metrics and test data, and the integration should be checked through functional testing. If the research later includes a controlled study of personalized educational content, that study would be needed to investigate effects on learning outcomes.

Overall, the framework defines how the three classification processes contribute to learner profiling while keeping the distinction between identifying learner characteristics and demonstrating the educational benefits of personalization.

3 RESEARCH METHODOLOGY

3.1 Research design

This research adopted a quantitative, web-based behavioral data collection approach to investigate the use of webcam-derived gaze information and mouse cursor interaction patterns for studying Visual–Verbal cognitive styles. A web-based learning module was developed to collect behavioral information while participants interacted with educational content presented in two different formats.

The module used a split-screen interface consisting of a visual panel and a text panel. Both panels presented the same educational content, but the visual panel included grayscale images, whereas the text panel presented the content without images. This design enabled the collection of interaction data from both content formats within the same learning environment.

The system collected behavioral information, including interaction time, scrolling activity, webcam-derived gaze estimates, and the participant's first interaction preference. The collected information was submitted to a backend service and stored for subsequent dataset preparation and research analysis.

The methodology consisted of six main stages:

1. Participant recruitment and consent.
2. Deployment and preparation of the web-based learning module.
3. Presentation of educational content through the split-screen interface.
4. Collection of cursor, scrolling, and webcam-derived gaze information.
5. Submission and storage of collected behavioral data through the backend.
6. Data preparation and organization for subsequent research analysis.

This methodology focused primarily on implementing the data collection environment and constructing a behavioral dataset. The collected dataset was intended to support subsequent investigation of relationships between observable interaction patterns and Visual–Verbal cognitive styles.

3.2 Data, participants or study materials

3.2.1 Participants

A total of 212 responses were initially collected from undergraduate students at the University of Kelaniya, Sri Lanka. The participants were from the Faculty of Science and the Faculty of Management and Information Technology.

Undergraduate students were selected as participants because the research focused on behavioral interactions within a web-based educational environment. Participants accessed the learning module and interacted with the educational materials according to the instructions provided during the data collection sessions.

A consent form was presented to participants before they took part in the activity. The purpose of the research and the nature of the activity should be described in the participant information provided for the study.

The collected responses were subsequently examined to identify incomplete records and sessions with insufficient recorded gaze interaction time. These checks were applied during dataset preparation to improve the suitability of the collected data for further analysis.

3.2.2 Split-Screen Learning Module

A web-based learning module was developed to collect behavioral data while participants interacted with educational content. The interface was designed with two parallel panels: a visual panel and a text panel.

The visual panel presented educational information supported by grayscale images. The text panel presented the same educational information without additional images. Using the same textual content helped maintain consistency between the two presentation formats while allowing the visual panel to provide additional visual material.

Each panel had an independent scrollbar. Therefore, participants could scroll through the visual and text content separately without requiring both panels to move together. This

arrangement allowed interaction activity to be recorded independently for each content format.

A button was placed at the bottom of each panel to allow participants to proceed after interacting with the relevant content. Participants could use the button associated with the panel they had completed without needing to scroll through the other panel to find its corresponding control.

When a participant clicked the relevant button, the frontend submitted the collected behavioral information to the backend service. The backend then stored the submitted data for subsequent research use.

The main purpose of this interface was to provide a consistent learning environment in which interactions with visual-supported and text-only content could be observed and recorded. The interface did not require participants to use specialized eye-tracking hardware. Instead, it used a standard webcam and software-based computer vision techniques to estimate gaze-related behavior.

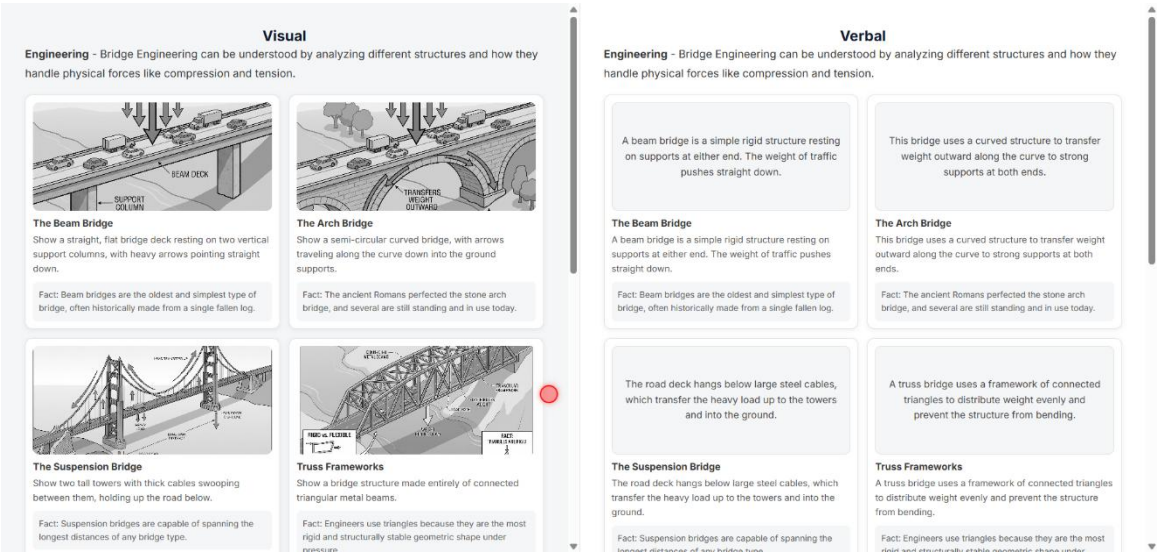


Figure 1 Split-screen learning module showing the visual and text content panels.

3.2.3 Cursor Interaction and Scrolling Data Collection

3.2.3 Cursor Interaction and Scrolling Data Collection

Mouse cursor interaction and scrolling information were collected as part of the behavioral data collection process. These measures were intended to represent how participants interacted with the visual and text panels during the learning activity.

The implementation recorded interaction time and scrolling activity separately for the two panels. The following variables were included in the submitted behavioral data.

Variable	Description
`visualTimeMs`	Recorded interaction time associated with the visual panel, measured in milliseconds.
`textTimeMs`	Recorded interaction time associated with the text panel, measured in milliseconds.
`visualScrolls`	Number of recorded scrolling actions associated with the visual panel.
`textScrolls`	Number of recorded scrolling actions associated with the text panel.

Table 2 Behavioral Data used

Recording the measures separately allowed the research dataset to distinguish between activity associated with the visual-supported content and activity associated with the text-only content.

The time-related variables were represented in milliseconds to provide a consistent numerical format for recording durations. The scrolling variables represented the number of recorded scrolling actions for each panel.

These measurements provided basic behavioral indicators of participants' interaction with the learning materials. However, interaction time and scrolling activity alone do not establish a participant's cognitive style. They were collected as observable behavioral measures for further analysis alongside the other recorded features.

3.2.4 Webcam-Based Gaze Data Collection

A standard webcam was used to collect information for software-based gaze estimation. The system used MediaPipe computer vision functionality to obtain face and eye-related information from the webcam input. These measurements were processed to estimate participants' gaze-related interaction with the visual and text panels.

The implementation included calibration to help account for differences between participants' face orientation and their estimated viewing direction. Calibration was important because head orientation alone does not necessarily represent the direction in which a participant is looking. The resulting measurements should therefore be interpreted as software-based gaze estimates rather than the precise measurements produced by dedicated eye-tracking equipment.

The following variables were included in the gaze-related data submitted to the backend.

Variable	Description
`visualGazeTimeMs`	Recorded gaze-related time associated with the visual panel, measured in milliseconds.
`textGazeTimeMs`	Recorded gaze-related time associated with the text panel, measured in milliseconds.
`totalActiveTimeMs`	Combined visual and text gaze-related time, calculated in milliseconds.

`firstInteractionPreference`	Records the first interaction preference identified by the implementation, with `NONE` used as the default when no preference is available.
`userId`	Identifies the participant or session associated with the submitted record.

Table 3 Model Variables

The `firstInteractionPreference` variable represented the initial preference recorded by the system. It was not treated as a definitive classification of cognitive style. Similarly, gaze-related duration was considered a behavioral measure rather than direct evidence of a participant's internal cognitive processes.

The use of webcam-based estimation provided a practical way to collect gaze-related information within a standard web-based learning environment. Nevertheless, the quality of these estimates could be influenced by factors such as webcam positioning, lighting conditions, face visibility, calibration, and individual differences.

3.2.5 Backend Data Transmission and Storage

A backend service was implemented to receive and store the behavioral data collected by the frontend. This separated the participant-facing learning interface from the data storage functionality and provided a centralized destination for submitted research records.

The web application was deployed using Microsoft Azure Container Apps so that it could be accessed during the university-based data collection sessions. Participants interacted with the deployed application through a web browser.

The frontend collected the relevant behavioral variables during the learning activity. When a participant clicked the button provided at the bottom of the relevant panel, the frontend submitted the collected information to the backend. The backend processed the incoming submission and stored the data for subsequent dataset preparation.

The submitted information included the participant or session identifier, panel-specific interaction times, scrolling counts, gaze-related durations, combined active time, and first interaction preference.

The identifier was obtained from the available user information when possible. When that information was unavailable, the implementation used the default identifier `session-demo-1`. Therefore, the identifier field should not automatically be interpreted as a unique, verified participant identity in every record. Records using a default identifier may require additional handling during dataset preparation.

The data transmission and storage process can be summarized as follows:

1. The participant interacts with the visual and text panels.
2. The frontend records the relevant behavioral measurements.
3. The participant clicks the appropriate panel button.
4. The frontend submits the collected measurements to the backend.
5. The backend receives and stores the submitted record.
6. The stored records are retrieved and organized for dataset preparation.

This process established a structured approach for transferring behavioral information from the learning interface to the backend. It also provided a consistent method for collecting records from multiple participants through the deployed web application.

3.3 Experimental or study procedure

The data collection procedure was conducted at the University of Kelaniya using the deployed web-based learning application. Two laptops were used during the data collection activity. Participants were provided with an explanation of the learning activity and instructions for interacting with the application.

The procedure consisted of the following steps.

Step 1: Participant introduction and consent

Participants were introduced to the research activity and provided with the consent form before participating. The data collection activity was conducted in accordance with the information and instructions provided to participants.

Step 2: Accessing the learning application

The web application was deployed on Microsoft Azure Container Apps and accessed through a web browser. This deployment made the learning module available through a web interface without requiring participants to install the application locally.

Step 3: Introduction to the learning interface

Participants were introduced to the split-screen learning module, which consisted of a visual panel and a text panel. They were informed how to interact with the content and use the buttons provided at the bottom of the panels.

Step 4: Interaction with educational content

Participants interacted with the educational material presented in the two panels. The visual panel contained grayscale images alongside the educational text, while the text panel presented the same textual content without images. Each panel could be scrolled independently.

Step 5: Collection of behavioral measurements

During the activity, the implementation collected the configured interaction-time and scrolling measures. Webcam-based processing was used to estimate gaze-related behavior, including the time associated with the visual and text panels. The system also recorded the first interaction preference when available.

Step 6: Submission of collected data

After interacting with the relevant content, participants used the button provided at the bottom of the appropriate panel. This action triggered the submission of the collected behavioral information to the backend service.

Step 7: Backend storage

The backend received and stored the submitted behavioral records. These records formed the initial dataset used in the subsequent data preparation process.

Step 8: Dataset preparation

The collected records were examined for missing values and insufficient gaze-related interaction time. Records that did not satisfy the defined data quality criteria were excluded from the final dataset.

This procedure provided a consistent process for collecting behavioral information from participants in a university learning environment. The use of the same web application and structured interaction process helped standardize the data collection activity.

3.4 Measures and evaluation metrics

3.4.1 Cursor and Scrolling Measures

The cursor and scrolling measures represented participants' interactions with the two content panels. The recorded interaction times and scrolling counts were maintained separately for the visual and text panels.

These measures were useful for examining differences in how participants interacted with the two content formats. For example, the recorded durations could be compared to investigate whether participants spent different amounts of time interacting with each panel. Similarly, scrolling counts could indicate differences in how participants navigated the content.

However, longer interaction time or a higher scrolling count does not necessarily indicate a particular cognitive style. Such measures can also be affected by reading speed, familiarity with the material, device usage, and other factors. Consequently, they were treated as behavioral observations rather than direct classifications.

3.4.2 Gaze-Related Measures

The gaze-related measures were derived from webcam-based computer vision processing. The system recorded estimated gaze-related time for the visual and text panels and calculated the combined time across both panels.

The separate panel measurements allowed the dataset to represent the distribution of recorded gaze-related activity between the two content formats. The combined duration provided an additional summary measure.

Because the system used a standard webcam and software-based estimation, the recorded measurements were subject to potential estimation errors. They were therefore interpreted as approximate behavioral measures and not as exact records of visual attention.

3.4.3 First Interaction Preference

The first interaction preference represented the initial content-panel preference recorded by the implementation. This variable was included to capture an early interaction pattern within the learning activity.

The value `NONE` was used when the implementation did not identify a first interaction preference. Such values were retained as a distinct state during data preparation rather than being interpreted as evidence of a visual or verbal preference.

The first interaction preference was considered alongside the other recorded measures. It was not used on its own to establish a participant's cognitive style.

3.4.4 Combined Behavioral Dataset

The behavioral measurements were organized into structured records containing the available participant or session identifier and the recorded interaction variables.

The main variables included in the dataset were:

Data category	Variables
Participant/session identification	`userId`
Interaction time	`visualTimeMs`, `textTimeMs`
Scrolling activity	`visualScrolls`, `textScrolls`
Gaze-related time	`visualGazeTimeMs`, `textGazeTimeMs`
Combined gaze-related time	`totalActiveTimeMs`
Initial interaction preference	`firstInteractionPreference`

The combined dataset provided a structured representation of the behavioral information collected through the learning module. It was prepared for subsequent research activities involving the investigation of behavioral patterns and Visual–Verbal cognitive styles.

3.5 Data analysis

Data preparation was conducted to improve the completeness and suitability of the collected behavioral records for further research analysis. The initial dataset contained 212 responses.

3.5.1 Data Quality Checks

The collected records were examined to identify missing values and sessions with insufficient recorded gaze-related interaction time. The following exclusion criteria were applied:

- **Missing values:** Records containing missing values in the required fields were excluded.
- **Insufficient gaze-related time:** Records with a total recorded gaze time below 10,000 milliseconds were excluded because they contained less than the minimum interaction time defined for the study.

The initial data cleaning process identified four records containing missing values and ten records with total gaze-related time below 10,000 milliseconds. Assuming these exclusion groups did not overlap, 198 records remained after the exclusions.

These criteria were used to reduce the influence of incomplete records and sessions with very limited gaze-related interaction. However, the threshold was a data preparation rule and did not establish that the remaining records represented accurate or complete measurements of participants' cognitive processes.

3.5.2 Dataset Organization

Following the data quality checks, the remaining records were organized using the defined behavioral variables. Panel-specific measures were retained separately so that interaction with visual-supported and text-only content could be distinguished.

The combined gaze-related time was also retained as a summary measure. Where applicable, the participant or session identifier was used to associate a record with the corresponding interaction session.

Missing or default identifiers required particular attention because the application could use `session-demo-1` when a user identifier was unavailable. Records sharing a default identifier should not be assumed to represent the same participant without additional evidence. Therefore, identifier consistency should be checked before conducting analyses that depend on individual-level records.

3.5.3 Preparation for Subsequent Analysis

The cleaned dataset was intended to support subsequent analysis of behavioral patterns associated with Visual–Verbal cognitive styles. The panel-specific time measures, scrolling activity, gaze-related measures, and first interaction preference provided different perspectives on participant interaction.

The prepared data could be examined to identify distributions, unusual values, differences between panels, and relationships among the recorded variables. Any later cognitive-style prediction analysis would also need to use the appropriate reference labels obtained through the standardized cognitive-style questionnaire.

The dataset preparation stage did not, by itself, demonstrate that the recorded behavioral measures could accurately predict cognitive styles. Such a conclusion would require a separate evaluation using an appropriate methodology.

3.6 Ethics, privacy, security and responsible AI

Ethical and privacy considerations were important because the research involved human participants and webcam-based behavioral data collection.

3.6.1 Participant Consent

A consent form was presented to participants before the data collection activity. Participants were provided with information about the activity and instructions for interacting with the learning module.

The final thesis should describe the actual consent procedure, including the information provided to participants and any applicable institutional approval, if obtained. Institutional approval should not be claimed unless it was formally granted.

3.6.2 Webcam-Based Data Collection

The system used a standard webcam to support software-based gaze estimation. Webcam processing can involve sensitive visual information, so the thesis should clearly state what the application processed and what information it retained.

In particular, the final description should confirm whether raw webcam frames or images were stored, or whether only derived gaze-related measurements were transmitted and retained. The system should not be described as storing only derived measurements unless this has been verified from the implementation.

3.6.3 Data Identification and Confidentiality

The collected records included a participant or session identifier. The use of identifiers required care to prevent records from being incorrectly associated with individuals.

The implementation's fallback identifier, `session-demo-1`, could be shared across multiple records when a user identifier was unavailable. Therefore, the identifier should not be treated as a reliable unique participant identifier in every case. Where participant-level analysis is required, records should be checked and linked using a suitable, consistently assigned identifier.

The final thesis should also explain the actual measures used to protect the stored data, restrict access, and handle any identifying information. Only safeguards that were implemented should be reported.

3.6.4 Responsible Interpretation of Behavioral Data

The collected interaction and gaze-related measures were treated as observable behavioral information. They were not assumed to provide direct or definitive evidence of a participant's cognitive processes.

Webcam-based gaze estimates can be affected by environmental conditions, calibration quality, camera positioning, and individual differences. Similarly, time spent on a content panel can reflect factors other than cognitive style. These limitations were considered when describing the collected data and its intended research use.

3.7 Reproducibility and research artefacts

The methodology was supported by a web-based learning module, a backend data collection service, a defined set of behavioral variables, and a data preparation procedure. Together, these elements established the process used to collect and organize the research dataset.

The main research artefacts included:

- The split-screen learning interface containing visual and text content panels.
- The independent scrolling functionality for the two panels.
- The frontend functionality for recording and submitting behavioral information.
- The webcam-based gaze estimation and calibration functionality.
- The backend service for receiving and storing submitted records.
- The deployed application hosted using Microsoft Azure Container Apps.
- The defined behavioral variables and data quality criteria.
- The cleaned dataset prepared for subsequent research analysis.

Documenting these artefacts helps explain how the data collection process was implemented. Reproducibility also depends on preserving the relevant software versions, configuration details, data collection instructions, variable definitions, and data cleaning rules.

The deployment environment made it possible to access the learning module through a web browser during the data collection sessions. However, reproducing the full environment would also require the relevant application source code, dependency information, deployment configuration, and any required backend settings, with secrets and sensitive participant information excluded from shared materials.

In summary, the methodology established a web-based environment for collecting behavioral data during interactions with visual-supported and text-only educational content. The implementation recorded interaction time, scrolling activity, webcam-derived gaze estimates, and first interaction preference, and submitted the collected information to a backend service for storage. The resulting dataset was then examined for missing values and insufficient gaze-related interaction time before being organized for subsequent

research analysis. This process provided the data collection foundation for investigating the relationship between behavioral interaction patterns and Visual–Verbal cognitive styles.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

This section describes the functional and non-functional requirements of the proposed Visual–Verbal Cognitive Style Prediction System. These requirements define the expected behavior of the system and the technical considerations that guide its design and implementation.

4.1.1 Functional Requirements

The functional requirements describe the main operations that the system must support.

FR1 – Learning Interface: The system shall provide a web-based learning interface containing separate panels for visual and verbal learning materials.

FR2 – Independent Content Navigation: The system shall allow learners to navigate the visual and verbal panels independently using separate scrolling controls.

FR3 – Webcam-Based Gaze Estimation: The system shall use MediaPipe FaceLandmarker to obtain facial and iris-related landmarks and estimate approximate gaze-related coordinates.

FR4 – Personalized Calibration: The system shall provide an initial webcam positioning check and a multi-stage calibration process to collect landmark observations and calculate personalized gaze-coordinate ranges.

FR5 – Behavioral Data Collection: The system shall collect the selected behavioral features, including image-associated cursor activity, scrolling activity, gaze-related behavior, and first interaction preference.

FR6 – Behavioral Data Submission and Storage: The system shall submit the collected behavioral observations to the backend for storage and subsequent research analysis.

FR7 – Dataset Preparation: The system shall support data cleaning, feature selection, categorical encoding, and numerical feature standardization before model training.

FR8 – Cognitive Style Classification: The system shall use a supervised machine learning model to predict visual–verbal cognitive style categories from the selected behavioral features.

FR9 – Model Evaluation: The system shall support evaluation using accuracy, precision, recall, F1-score, weighted F1-score, and a confusion matrix.

FR10 – Reproducible Prediction Processing: The system shall preserve the feature representation and preprocessing transformations required to use the trained model consistently.

4.1.2 Non-Functional Requirements

The non-functional requirements describe the quality attributes and constraints considered during system design.

NFR1 – Usability: The learning interface should allow participants to interact with visual and verbal content without unnecessary complexity.

NFR2 – Compatibility: The system should operate in a supported web browser with access to a compatible webcam for gaze-related data collection.

NFR3 – Data Quality: The system should identify records that do not meet the specified data-quality criteria before using them for model development.

NFR4 – Maintainability: The learning interface, gaze-estimation component, backend, data-processing workflow, and machine learning model should be organized as distinct components where practical.

NFR5 – Reproducibility: The model-development workflow should use documented preprocessing steps and consistent data-splitting and evaluation procedures.

NFR6 – Privacy: The system should limit collected and processed information to what is necessary for the research and follow the applicable data-protection requirements.

NFR7 – Responsible Prediction: The system should treat predictions as estimates of questionnaire-derived categories rather than definitive measurements of a learner's cognitive characteristics.

NFR8 – Reliability of Gaze-Related Features: The system should provide positioning checks and personalized calibration to address some sources of inconsistency in webcam-based gaze estimation.

These requirements provide the basis for the architectural decisions and detailed design described in the following sections.

4.2 Component Architecture Within the Complete Solution

4.2.1 Overview of the Architecture

The proposed system follows a modular web-based architecture that separates the learning interface, gaze estimation, behavioral data collection, backend storage, dataset preparation, and machine learning operations.

The learner interacts with the system through a React-based web interface developed using Vite. The interface presents visual and verbal learning materials in separate panels and collects behavioral observations during the learning activity.

MediaPipe FaceLandmarker processes webcam observations to identify facial and iris-related landmarks. A personalized calibration component checks the participant's positioning and collects additional landmark observations at predefined targets. The resulting personalized coordinate ranges are stored in browser local storage.

The behavioral collection component combines the selected interaction information into structured records. These records are submitted to the backend and stored for research purposes. The resulting dataset is processed to prepare the input features and questionnaire-derived target labels required for supervised machine learning.

An SVM classifier with an RBF kernel is used to predict the visual–verbal cognitive style category associated with the observed behavioral features.

4.2.2 Main Architectural Components

The architecture consists of the following components.

1. Web-Based Learning Interface

The React and Vite frontend provides the learning environment. It displays grayscale images in the visual panel and text-only content in the verbal panel. Independent scrollbars and submission controls support the collection of panel-associated interaction data.

2. Gaze Estimation and Calibration

The gaze-estimation component uses MediaPipe FaceLandmarker to obtain facial and iris-related landmarks. The calibration component checks face positioning and collects observations while participants follow targets involving iris movement, head movement, and refinement. It calculates personalized minimum and maximum gaze-coordinate ranges for subsequent processing.

3. Behavioral Data Collection

This component collects cursor activity, scrolling activity, gaze-related information, and first interaction preference. The observations are organized into the behavioral features used in the research dataset.

4. Backend and Data Storage

The backend receives the behavioral records submitted by the frontend and stores information needed for dataset preparation and analysis. The stored records support the association between behavioral features and questionnaire-derived cognitive style labels.

5. Dataset Preparation

The data-processing workflow removes records that fail the specified data-quality checks, separates input features from target labels, encodes categorical variables, standardizes numerical features, and addresses class imbalance within the training data.

6. Machine Learning and Evaluation

The SVM classifier learns relationships between the behavioral input features and the questionnaire-derived target categories. Its performance is assessed using a held-out test dataset and the selected classification metrics.

4.2.3 Component Interaction and Data Flow

The workflow begins when a participant enters the learning environment. The participant completes the required initial research questionnaires and proceeds to the learning module.

The application checks webcam positioning and guides the participant through the calibration process. Once calibration is completed, the participant interacts with the visual and verbal learning materials while the system collects behavioral observations.

The collected information is submitted to the backend and stored for research analysis. During dataset preparation, records that fail the defined data-quality checks are excluded, and the selected behavioral features are separated from the questionnaire-derived target labels.

The processed dataset is divided into training and test subsets. The SVM classifier is trained using the training data, while the held-out test data are used to assess predictive performance.

This component-based design separates data acquisition from model development and evaluation. It also allows gaze-estimation limitations to be investigated separately from the classification performance of the machine learning model.

4.3 Design Alternatives and Decisions

This section explains the principal design decisions made during the development of the proposed system and the reasons for selecting the implemented approaches.

4.3.1 Webcam-Based Gaze Estimation

An important design decision concerned how to obtain gaze-related information during the learning activity. The selected approach uses MediaPipe FaceLandmarker to estimate gaze-related coordinates from facial and iris landmarks captured through an ordinary webcam.

This approach provides a webcam-based alternative to using a dedicated eye-tracking device. It also integrates with the web-based learning interface and supports the collection of behavioral information during the learning activity.

However, the initial implementation produced inconsistent gaze estimates under the observed conditions. Investigation highlighted that camera positioning, learner-to-camera distance, head position, lighting, and individual facial characteristics can affect the detected landmarks.

Consequently, the system was extended with a personalized calibration process. Rather than relying only on the initial landmark estimates, the calibration component checks face positioning and collects observations at predefined screen targets.

The decision to retain MediaPipe and introduce calibration was based on the need to address positioning-related inconsistencies within the existing web-based implementation. The calibration process is intended to improve consistency, but its actual accuracy improvement requires separate experimental validation.

4.3.2 Personalized Calibration Strategy

The calibration design considered the need to obtain individualized observations under the participant's current camera and positioning conditions.

The implemented approach uses multiple stages: initial positioning, iris-related observation collection, head-movement observation collection, and refinement. At the end of calibration, personalized minimum and maximum gaze-coordinate values are calculated and stored in browser local storage.

This approach was selected because it collects observations across several target positions and considers changes in both eye-related and facial landmarks.

An important limitation is that the resulting values represent coordinate ranges rather than a fully validated mapping between gaze estimates and exact screen positions. Therefore, the design provides personalized reference bounds but does not guarantee accurate screen-point tracking.

4.3.3 Split-Screen Learning Interface

The system uses a split-screen interface containing separate visual and verbal content panels. The visual panel presents grayscale images, while the verbal panel presents text without images.

An alternative would be to present the content sequentially or combine images and text in the same panel. However, the split-screen approach was selected to allow learners to interact with both content types independently and to support the collection of panel-associated cursor, scrolling, and gaze-related information.

Independent scrollbars allow the two panels to be navigated separately. The first interaction preference also provides an indication of which content type the participant interacts with first.

The design provides a structured setting for investigating relationships between learner interaction and questionnaire-derived visual–verbal cognitive style categories. Nevertheless, interaction preference alone does not establish a learner's cognitive style.

4.3.4 Behavioral Feature Selection

The system uses a selected group of behavioral features rather than including every available field in the dataset.

The selected features represent cursor activity, scrolling activity, gaze-related behavior, and first interaction preference. These were chosen to capture different aspects of interaction with the visual and verbal learning materials.

Fields such as `UserID` and `Row Note` are excluded from model inputs because they are not intended to represent the behavioral patterns under investigation. The duration fields `cursorScrollTimeMs` and `gazeTimeMs` are also excluded according to the selected feature design.

This approach keeps the model input focused on the chosen behavioral indicators and reduces the risk of learning associations based on identifiers or unrelated metadata.

4.3.5 Selection of the SVM Classifier

The system uses a Support Vector Machine with an RBF kernel to classify the questionnaire-derived visual–verbal cognitive style categories.

The RBF kernel was selected to allow the model to represent nonlinear relationships between behavioral features and target categories. Feature standardization was included because differences in numerical feature scales can affect SVM classification.

Class imbalance was addressed using balanced class weights and SMOTE applied to the training subset. The model was selected because it performed best in the comparison conducted during the research.

The final justification for this selection should be supported by the actual comparative evaluation results reported in Chapter 6. The classifier's performance should be interpreted in relation to the questionnaire-derived labels and the size and composition of the available dataset.

4.4 Detailed Design

4.4.1 Learning Interface Design

The learning interface is implemented using React and Vite. It presents two separate panels for the learning activity: a grayscale image panel and a text-only panel.

Each panel has its own scrollbar, allowing participants to navigate the content independently. Submission controls are provided to send the collected behavioral observations to the backend.

The split-screen layout is designed to support the observation of interaction patterns associated with visual and verbal content.

4.4.2 Webcam Positioning and Personalized Calibration

The calibration component is implemented in the React application using `react-webcam` and MediaPipe FaceLandmarker.

The first stage checks whether a face has been detected and whether the participant is approximately centered within the webcam frame. The component uses facial landmark boundaries to estimate the center and width of the face.

The current implementation accepts horizontal and vertical face-center coordinates between 0.35 and 0.65. It also considers the normalized face width suitable when it is greater than 0.25 and less than 0.55. These are implementation-specific normalized thresholds and do not represent exact physical distances.

After the positioning check, the calibration component guides the participant through the following stages:

1. **Iris-related observations:** The participant follows targets across central, intermediate, and near-edge screen positions while attempting to keep the head relatively still.
2. **Head-movement observations:** The participant follows directional targets involving upward, downward, leftward, and rightward movements, with center observations between directions.
3. **Refinement observations:** The participant follows a smaller set of targets around the center and quarter positions of the screen.

The component processes the captured landmarks to obtain approximate horizontal and vertical gaze-related coordinates. It calculates the minimum and maximum observed values for each coordinate and saves the personalized ranges in browser local storage under the key `customGazeRules`.

These values are used as individualized reference information. The calibration process does not independently verify the accuracy of the estimated gaze coordinates.

4.4.3 Gaze-Related Feature Extraction

The gaze-estimation component uses facial and iris-related landmarks to derive approximate gaze coordinates.

The position of the nose relative to the detected facial region is used to estimate horizontal and vertical head-pose information. The component also identifies the left and right iris regions and estimates iris offsets relative to eye-related reference positions.

The left and right eye observations are combined to calculate approximate horizontal and vertical gaze coordinates. The implementation applies bounded transformations, sensitivity adjustments, and nonlinear adjustments to the iris-offset values before combining them with head-pose information.

The resulting coordinates provide intermediate information for gaze-related behavioral processing. The model input feature `ImageGazeRatio` is a summarized behavioral indicator associated with image content and should not be confused with the raw gaze coordinates.

Because the system uses webcam-based landmark estimation, the extracted gaze-related features remain approximate and may be influenced by positioning, lighting, head movement, and landmark-detection quality.

4.4.4 Behavioral Feature Design

The system uses four principal behavioral features.

Feature	Description	Purpose
`imageCursorRatio`	Relative cursor activity associated with image-based content.	Represents interaction associated with visual material.
`imageScrollRatio`	Relative scrolling activity associated with image-based content.	Represents navigation associated with visual material.

`ImageGazeRatio`	Gaze-related ratio associated with image content.	Represents estimated gaze behavior associated with visual material.
`FirstInteractionPreference`	Categorical indication of the first content type selected, such as TEXT or VISUAL.	Represents initial interaction preference.

The features capture different aspects of behavior during the learning activity. Cursor and scrolling features describe interface interactions, while the gaze-related feature depends on approximate webcam-based estimates. First interaction preference represents the initial content choice.

These features do not independently measure comprehension, learning effectiveness, or a permanent cognitive characteristic. Their purpose is to investigate whether observable interaction patterns are associated with questionnaire-derived visual–verbal categories.

The exact calculation of each ratio follows the implemented feature-extraction process. The feature names alone should not be treated as a complete mathematical definition.

4.4.5 Dataset Preparation

The initial dataset contained 212 records. Data cleaning identifies records with missing values in required gaze-related or target-label fields and records with `gazeTimeMs` below 10,000 milliseconds.

The final dataset size is determined after applying these filters and checking for overlapping exclusions.

The target variable is `Questionnaire Label (Cognitive Style)`. The model input excludes `Row Note`, `UserID`, `cursorScrollTimeMs`, `gazeTimeMs`, and the target label itself.

`FirstInteractionPreference` is transformed using one-hot encoding with the first category dropped. The target categories are encoded numerically using label encoding.

The dataset is split into training and test subsets using an 80:20 stratified split with a fixed random state of 42. Numerical input features are standardized using `StandardScaler`, which is fitted on the training subset and then applied to the test subset.

SMOTE is applied only to the training data to address class imbalance. The original test data remain unchanged for evaluation.

These steps establish a consistent preprocessing workflow while reducing the risk of information leakage from the test subset into model training.

4.4.6 SVM Model Design

The SVM classifier uses an RBF kernel to predict the questionnaire-derived visual–verbal cognitive style category.

The model configuration uses a regularization parameter of 2.0, a gamma setting of `scale`, balanced class weights, probability estimation, and a fixed random state of 42.

Feature standardization is applied because SVM classification can be affected by differences in numerical feature scales. Balanced class weights adjust the relative importance of classification errors across classes, while SMOTE creates synthetic minority-class examples in the training subset.

The model is trained using the selected behavioral features and the questionnaire-derived target labels. During prediction, incoming data must follow the same feature representation, encoding scheme, and feature order used during training.

The resulting output is a predicted category associated with the behavioral input. It should be interpreted as an estimate learned from the available research data rather than a definitive assessment of cognitive style.

4.4.7 Model Evaluation

The trained classifier is evaluated using the held-out test subset. The evaluation considers accuracy, class-level precision, recall, F1-score, weighted F1-score, and a confusion matrix.

Accuracy summarizes the proportion of correctly classified observations. Precision and recall provide class-specific information, while F1-score combines these measures. Weighted F1-score summarizes class-level F1-scores according to the number of actual observations in each class. The confusion matrix shows the distribution of correct and incorrect predictions across classes.

Class-level measures are important because overall accuracy may conceal weak performance for a minority class.

The evaluation of cognitive style classification is separate from the evaluation of gaze-estimation accuracy. To establish whether personalized calibration improves gaze estimation, calibrated and uncalibrated estimates would need to be compared against known target positions under comparable conditions.

The actual classification metrics and confusion matrix are reported in Chapter 6.

4.5 Standards and Professional Practice

4.5.1 Ethical Data Collection

The research data were collected at the University of Kelaniya from undergraduate participants belonging to the Faculty of Science and the Faculty of Management and IT, using two laptops.

The learning activity collects behavioral observations during interaction with visual and verbal learning materials. The collected information includes cursor activity, scrolling activity, gaze-related information, first interaction preference, and questionnaire-derived labels.

Participants should be informed about the purpose of the research, the information being collected, and the intended use of the data. Any consent procedures and institutional approvals should be described according to the procedures actually followed during the study.

4.5.2 Privacy and Data Protection

The system uses webcam observations to obtain facial and iris-related landmarks. Since webcam observations can contain sensitive visual information, collection and processing should be limited to what is necessary for the research objectives.

The field `UserID` is excluded from the selected model inputs. However, excluding an identifier from model inputs does not automatically make the entire dataset anonymous.

The calibration component stores personalized gaze-coordinate ranges in browser local storage under `customGazeRules`. Storage, access, retention, and deletion procedures for other research data should follow the actual implementation and applicable institutional requirements.

Claims that raw webcam images are discarded or that all collected information is anonymous should only be included if these behaviors have been verified.

4.5.3 Responsible Use of Machine Learning

The classifier learns relationships between observed behavioral features and questionnaire-derived target labels. The quality of its predictions depends on the dataset, the target labels, the selected features, and the evaluation procedure.

Learner behavior may be influenced by reading speed, interface familiarity, environmental conditions, task difficulty, and individual interaction habits. Therefore, a predicted category should not be interpreted as an independently verified or permanent characteristic.

The system is intended for research into behavioral patterns associated with visual–verbal cognitive style. Its predictions should not be used as the sole basis for high-impact educational decisions or to restrict learners' access to particular learning materials.

4.5.4 Reproducibility and Evaluation Integrity

The data-processing workflow uses documented feature exclusions, categorical encoding, an 80:20 stratified train–test split, and a fixed random state of 42.

The scaler is fitted using the training data and applied to the test data. SMOTE is also restricted to the training subset, ensuring that synthetic observations are not introduced into the held-out test data.

The same feature representation, encoding scheme, scaling transformation, and feature order must be maintained between training and prediction.

The final dataset size, class distribution, and model metrics should be reported using the actual experimental outputs. Model comparisons should use consistent evaluation procedures to support a fair comparison of results.

4.5.5 Maintainability and Professional Practice

The system separates the learning interface, gaze-estimation component, backend, dataset-preparation workflow, and machine learning model into distinct components. This organization supports maintenance and allows individual components to be updated without necessarily redesigning the entire system.

Future improvements may include more rigorous gaze-estimation validation, evaluation under different camera conditions, larger participant samples, and comparisons with alternative machine learning models.

Professional practice also requires accurate reporting of system limitations. Personalized calibration is intended to address gaze-estimation inconsistency, but an improvement in accuracy should not be claimed without supporting measurements. Similarly, model predictions should be interpreted in relation to the questionnaire-derived labels rather than presented as definitive measurements of a learner's cognitive characteristics.

4.5.6 Chapter Summary

This chapter described the requirements, architecture, design alternatives, detailed implementation design, and professional considerations of the Visual–Verbal Cognitive Style Prediction System.

The architecture combines a React-based split-screen learning interface, MediaPipe-based gaze estimation, personalized calibration, behavioral data collection, backend storage,

dataset preprocessing, and an SVM classifier. The design decisions address the need to collect behavioral information associated with visual and verbal content while acknowledging the limitations of webcam-based gaze estimation.

The detailed design established the behavioral features, data-cleaning process, train–test split, standardization, training-only SMOTE, SVM configuration, and evaluation measures. Ethical data collection, privacy, responsible interpretation, and reproducibility were also considered.

These design decisions provide the foundation for the implementation, integration, testing, and results discussed in the following chapters.

5 Implementation, Integration and Testing

5.1 Development Environment and Technology Stack

5.1.1 Development Environment

The Visual–Verbal Cognitive Style Prediction component was developed as a web-based application that combines educational content presentation, behavioral data collection, and machine learning-based classification. The development environment supported the implementation of the frontend interface, backend processing, data storage, webcam-based gaze estimation, and model training. The system was developed and tested using the tools required for these different responsibilities. The learning module was made accessible through a web browser, allowing participants to interact with the learning materials using laptop computers. The research data collection environment was hosted using Azure Container Apps.

Component	Technology or environment	Purpose
Frontend	React.js	Implements the interactive learning interface
Frontend development tool	Vite	Supports frontend development and building
Backend	Node.js	Handles backend processing and behavioral data submission
Gaze estimation	MediaPipe-based facial landmark and iris information	Estimates visual attention using webcam input
Machine learning	Python and scikit-learn	Supports data preprocessing, algorithm comparison, and classification
Data preprocessing	Pandas and NumPy, where used in the implementation	Supports dataset preparation and numerical processing
Model balancing	SMOTE	Addresses class imbalance during training

Data storage	Mongo DB	Stores behavioral records
Deployment environment	Azure Container Apps	Hosts the web application and relevant services

Table 4 Development environment

5.1.2 Frontend Implementation Environment

The frontend was implemented using React.js, with Vite supporting the development environment. The interface was designed to display the image-based and text-based learning materials side by side. The frontend also supports interaction with the learning content, independent scrolling, and submission of behavioral information when the learner completes the relevant interaction stage. This approach separates the user interface from the backend data-processing responsibilities. It also makes the learning interface easier to maintain and extend.

5.1.3 Backend and Data Processing Environment

The backend receives behavioral information collected through the learning interface and associated data collection components. The submitted records are processed and stored for subsequent analysis. The machine learning workflow was implemented using Python-based data processing and classification tools. The workflow includes data cleaning, categorical feature preparation, dataset splitting, model training, and performance evaluation. The backend data collection process and the model-development process serve different purposes. The backend supports the collection and persistence of behavioral observations, while the model-development workflow uses prepared records to train and evaluate the classifier.

5.1.4 Deployment Environment

The application was hosted using Azure Container Apps to provide access to the web-based learning module during research data collection. The deployment environment allowed participants to access the application through their browsers without requiring the entire development environment to be installed on each laptop.

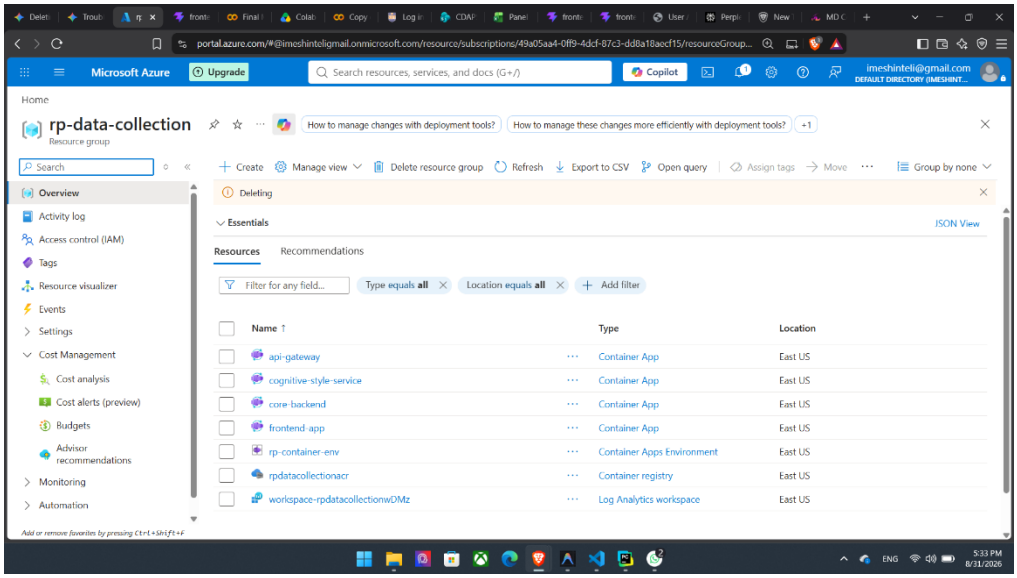


Figure 2 Deployed learning module Azure console microservices

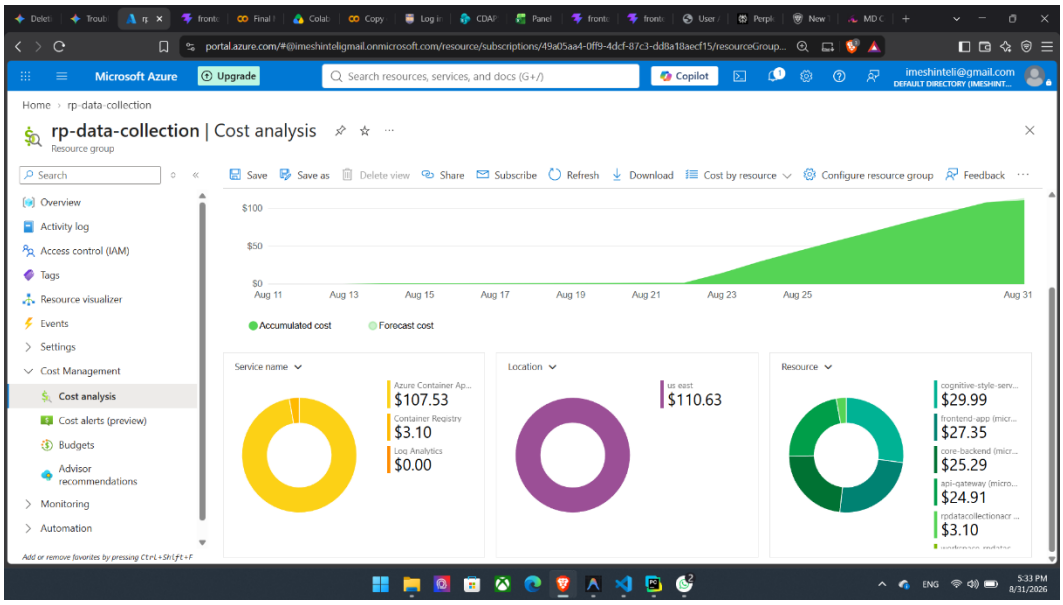


Figure 3 Deployed learning module Azure console costs

5.2 Component Implementation

5.2.1 Learning Module and Behavioral Data Collection

The learning module was implemented as a split-screen interface containing image-based and text-based learning materials. The image panel displays grayscale images, while the text panel provides the corresponding text-based content without images. Each panel has an independent scrollbar. This allows learners to navigate the two content formats separately and provides an opportunity to observe differences in their interaction

behavior. Buttons at the bottom of the panels allow learners to complete the relevant interaction stage. When a button is activated, the collected behavioral information is submitted to the backend for processing and storage. The behavioral collection process combines different sources of information. Cursor movement and scrolling provide information about interaction with the displayed material, while webcam-based gaze estimation provides an approximate indication of visual attention. Combining these observations enables the system to represent learner interaction through several behavioral features instead of depending on a single measurement.

5.2.2 Behavioral Ratio Calculation

The implementation uses ratio-based features to represent relative patterns of interaction with the learning materials. These features form part of the input dataset used for machine learning. The main behavioral features include imageCursorRatio, imageScrollRatio, and ImageGazeRatio.

The image cursor ratio represents cursor activity associated with the image-based content. It provides an indication of how cursor interaction is distributed in relation to the visual material. The image scroll ratio represents scrolling activity associated with the image-based content. It provides information about how the learner navigates the image panel relative to the relevant interaction measurements. The image gaze ratio represents gaze-related activity associated with the image content. This feature is derived from webcam-based gaze estimation and should be interpreted in light of the limitations of webcam-based measurements. The implementation also includes FirstInteractionPreference, which represents the learner's initial interaction choice, such as selecting TEXT or VISUAL content. This feature provides categorical information about the learner's initial preference before the other behavioral observations are considered.

Feature	**Data type**	**Role in prediction**
imageCursorRatio	Numerical	Represents relative cursor activity associated with images
imageScrollRatio	Numerical	Represents relative scrolling activity associated with images

ImageGazeRatio	Numerical	Represents the gaze-related ratio associated with images
FirstInteractionPreference	Categorical	Represents the initial content-format choice

Table 5 Behavioral features used in the implementation

The ratios provide a compact representation of behavioral activity. However, the exact mathematical definition of each feature should be documented using the actual implementation, particularly the numerator, denominator, and treatment of missing or zero-duration observations. The calculated features are stored or prepared as part of the dataset used for subsequent model development. Their usefulness is then evaluated through classification performance rather than assumed from their definitions alone.

5.2.3 Dataset Preprocessing and Feature Preparation

The collected dataset was prepared before training the machine learning model. The purpose of preprocessing was to improve data quality, establish a consistent feature representation, and ensure that the model received the expected inputs. The preprocessing process included the following steps:

1. Missing-value handling: Records with missing values in the important gaze-related feature or questionnaire-derived target label were excluded from the model-development dataset. This ensured that the required input and target information was available for the affected training records.
2. Filtering insufficient gaze duration: Records with a total gaze duration below 10,000 milliseconds were excluded. This criterion was used to remove sessions that did not meet the minimum duration requirement established for the experiment. This filtering rule represents a data-quality decision. It does not independently prove that the remaining gaze measurements are accurate.
3. Exclusion of identifiers and non-predictive fields: Fields such as participant identifiers, row notes, and designated timing variables were excluded from the model input features. These fields may be useful for managing or filtering the dataset, but they were not intended to serve as predictive behavioral inputs.
4. Categorical feature encoding: The initial interaction preference was converted into a numerical representation using one-hot encoding. This allowed the categorical preference to be included in the classification process.

5. Target-label encoding: The questionnaire-derived cognitive style labels were converted into numerical labels for model training. The original category mapping was retained so that model predictions could be interpreted in terms of the intended cognitive style categories.
6. Dataset splitting: The prepared dataset was divided into training and testing subsets using an 80:20 split. Stratification was applied to preserve the distribution of the target classes as far as possible across the two subsets. The random state was fixed at 42 to support reproducibility of the configured random operations.

Stage	**Implemented approach**
Missing data	Exclude records missing required gaze or target-label information
Gaze duration	Exclude records below 10,000 ms
Input preparation	Exclude identifiers, notes, target labels, and designated timing fields
Categorical inputs	One-hot encode the initial interaction preference
Target preparation	Encode questionnaire-derived cognitive style labels
Dataset split	80% training and 20% testing
Class distribution	Stratified splitting
Reproducibility	Fixed random state of 42

Table 6 Dataset preparation summary

The final number of records should be confirmed from the actual cleaned dataset. The initial dataset contained 212 records, while the described exclusions involved four records with missing values and ten records with insufficient gaze duration. If these exclusion groups do not overlap, the final dataset contains 198 records. If some records appear in both groups, the final count will be different. Therefore, the final thesis should report the count directly from the cleaned dataset rather than assuming that the exclusions are disjoint.

5.2.4 Machine Learning Algorithm Comparison and SVM Implementation

Several classification algorithms were evaluated to identify a suitable model for predicting Visual–Verbal cognitive style from the prepared behavioral features. The comparison was conducted using the same general prediction task and held-out evaluation

approach. The purpose was to determine which candidate provided the most suitable performance on the available dataset.

The Support Vector Machine was selected because it achieved the best performance in the actual algorithm comparison performed for this component. The selected configuration used an RBF kernel, a regularization parameter of 2.0, and the scale-based gamma setting. Feature standardization was included because the numerical ranges of the behavioral features may differ. Without appropriate scaling, features with larger numerical values can have a disproportionate influence on an SVM classifier. The implementation also incorporated class balancing. Balanced class weights were used in the SVM configuration, while SMOTE was included in the training workflow to address class imbalance by generating synthetic samples for underrepresented classes. Resampling must be restricted to the training data. The test data must remain separate from model fitting and resampling so that evaluation reflects performance on unseen records.

The selected model was configured to provide probability estimates alongside its predicted category. These estimates can indicate the model's relative support for different categories, although their calibration should not be assumed without a separate assessment. The training workflow produces a fitted model that can be used to classify new behavioral feature records. To ensure consistent predictions, operational input records must use the same feature definitions, categorical encoding, column order, and preprocessing transformations as the training data.

5.2.5 Model Evaluation Implementation

The model evaluation process used multiple classification metrics to assess predictive performance. Accuracy measured the overall proportion of correctly classified test records. Precision and recall provided class-specific information about prediction quality, while the F1-score combined these two measures. Weighted F1-score provided an aggregate measure that accounted for the number of test records in each class. The confusion matrix was used to examine the relationship between actual questionnaire-derived categories and predicted categories. This enabled the identification of classes that were correctly recognized and classes that were more frequently confused with one another. The same evaluation criteria should be applied consistently across the compared

algorithms. The selected SVM results should be reported using the actual evaluation output generated from the held-out test set.

Measure	Purpose
Accuracy	Measures overall classification correctness
Precision	Measures the correctness of predictions for each class
Recall	Measures the proportion of actual class records identified correctly
F1-score	Balances precision and recall
Weighted F1-score	Aggregates class-level F1-scores while accounting for class support
Confusion matrix	Shows actual versus predicted class assignments

Table 7 Evaluation Metrices

5.3 Integration with the Other Components

5.3.1 Integration with the Wider Learner Profile

The Visual–Verbal prediction component is designed to contribute one dimension to the wider learner profiling system. The broader system includes questionnaire-based assessment of learning approaches through the ASSIST questionnaire and assessment of Analytic–Wholistic cognitive style through a separate questionnaire. Visual–Verbal cognitive style is handled differently because the operational prediction is based on behavioral information collected while the learner interacts with educational content. This separation allows the system to combine information obtained through questionnaires with information inferred from observed interaction.

5.3.2 Questionnaire Workflow

During the intended first-login workflow, learners complete the ASSIST and Analytic–Wholistic questionnaires to establish the relevant parts of their learner profile. The Visual–Verbal model follows a different process. Questionnaire-derived Visual–Verbal labels are used during research and model development to train and evaluate the classifier. After the model has been trained, behavioral information collected during learning interactions can be used to estimate the learner's Visual–Verbal category. This approach is intended to reduce the need for learners to complete a Visual–Verbal

questionnaire during every learning session. It does not eliminate the need for questionnaire-based labels in model development or subsequent validation.

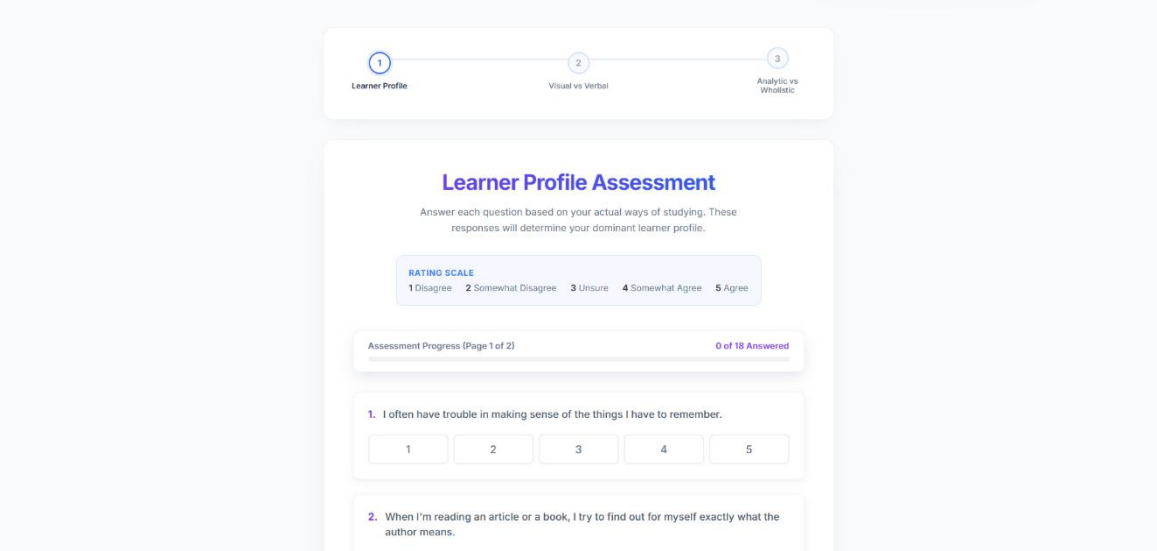


Figure 4 Integration of the Visual–Verbal component with the learner profile.

5.3.3 Behavioral Data Integration

The frontend, backend, and machine learning workflow have different responsibilities. The frontend presents the learning content and initiates the submission of behavioral information. The backend receives and stores the relevant observations. The model-development workflow prepares the collected records and uses them to train and evaluate the classifier. For operational prediction, the behavioral features supplied to the model must match the features used during training. Any mismatch in feature names, encoding, column order, or preprocessing can lead to incorrect predictions or inference errors. The integration design therefore depends on maintaining a consistent definition of the input features across collection, storage, preprocessing, training, and prediction.

5.3.4 Integration Considerations

The following considerations are important when integrating the component with the wider platform:

- Behavioral records must be associated with the appropriate learner or session.
- Required features must be available before prediction is attempted.

- Missing or invalid input values must be handled according to the implemented validation rules.
- The model's output categories must correspond to the categories expected by the learner profile.
- The component must not treat a model prediction as equivalent to a directly observed or independently verified cognitive characteristic.
- The integration process should be tested using actual requests and responses before claiming that end-to-end operation has been verified.

The thesis should distinguish between the intended integration design and integration functionality that has been demonstrated through actual testing.

5.4 Verification and Software Testing

5.4.1 Testing Strategy

Testing was intended to examine the behavior of the learning interface, the collection of behavioral information, the preparation of the dataset, the model training process, and the prediction workflow. Different testing levels are relevant because a component can work correctly in isolation but still experience problems when integrated with other services. The test cases below provide a structure for documenting verification. Their status must be updated according to the tests that were actually performed.

5.4.2 Functional Testing

Functional testing examines whether each major feature behaves according to its requirements.

Test ID	**Test case**	**Expected result**	**Actual result/status**
FT01	Open the learning module	The learning interface loads correctly	PASS
FT02	Display image and text panels	Both content formats are displayed	PASS
FT03	Scroll each panel independently	Scrolling one panel does not unintentionally control the other	PASS

FT04	Complete the interaction stage	The relevant button triggers the intended action	PASS
FT05	Submit behavioral records	The backend receives the expected data	PASS
FT06	Store collected records	The records are persisted in the configured storage system	PASS
FT07	Estimate gaze information	The gaze component produces the intended estimated information under supported conditions	PASS
FT08	Prepare the model dataset	Required preprocessing operations are applied	PASS
FT09	Generate a model prediction	The trained model returns an output for valid input features	PASS
FT10	Handle incomplete input	Invalid or incomplete inputs are handled according to the implemented validation rules	PASS

Table 8 Functional test cases

These test cases describe the expected verification activities. A test should be marked as passed only when its expected result has been observed and recorded.

5.4.3 Data Quality Testing

Data quality testing examines whether the dataset satisfies the requirements established for model development. The relevant checks include:

- Identifying missing values in required features and target labels.
- Confirming that records below the defined gaze-duration threshold are excluded.
- Checking that categorical values are represented consistently.
- Verifying that the target labels belong to the expected categories.
- Confirming that excluded identifiers and designated timing fields do not enter the model input.
- Checking the number of records remaining after preprocessing.
- Confirming that class balancing is applied only to the training data.

The final report should include the actual number of records before and after preprocessing, the class distribution, and any additional data-quality issues identified during the experiment.

5.4.4 Machine Learning Testing

Machine learning testing focuses on the training process and predictive behavior of the selected classifier. The test activities should include confirming that the training process completes successfully, that the trained model produces predictions for valid input records, and that the required evaluation metrics can be generated. The classification results should be checked against the held-out test labels. The confusion matrix should also be reviewed to identify any systematic misclassification between the categories. Where the model is compared with alternative algorithms, the same data split and evaluation metrics should be used to support a fair comparison.

5.4.5 Integration Testing

Integration testing examines whether the frontend, backend, data storage, and prediction components exchange information correctly. Relevant integration tests include submitting behavioral records from the learning interface, confirming that records are stored with the required fields, preparing the stored data for model development, and verifying that a prediction can be generated using the expected feature representation. If the trained model is connected to the operational platform, an additional test should verify that the prediction can be returned to the component that consumes the learner profile.

Test ID	**Integration scenario**	**Expected result**	**Actual result/status**
IT01	Frontend to backend	Behavioral data is received in the expected format	PASS
IT02	Backend to storage	Valid behavioral records are stored correctly	PASS
IT03	Stored dataset to preprocessing	Records are transformed according to the defined rules	PASS
IT04	Preprocessing to trained model	Input features match the model's expected representation	PASS

IT05	Model to prediction consumer	The predicted category is returned in the expected format	PASS
IT06	Component to learner profile	The prediction is available to the wider system	PASS

Table 9 Integration test cases

If the model was evaluated offline and was not connected to the live learner profile, the operational integration tests should be marked as not implemented or not applicable, as appropriate. Offline model evaluation should not be presented as proof of live end-to-end integration.

5.4.6 Testing Limitations

The interpretation of testing results must account for the size and composition of the collected dataset, the limitations of webcam-based gaze estimation, and the fact that the target labels originate from questionnaire responses. A successful classification result does not establish that the model can identify a universal or permanent cognitive style. It indicates how well the model reproduces the questionnaire-derived categories under the conditions of the evaluation. Further testing with additional participants, different learning materials, and varied environmental conditions would help assess the model's generalizability.

5.5 Deployment and Operation

5.5.1 Deployment

The web-based learning application was hosted using Azure Container Apps for research data collection. The deployment enabled participants to access the learning module through a browser on the available laptop devices. The deployed application supported the presentation of learning content and the collection of relevant behavioral information. The machine learning model-development workflow was used to prepare the collected dataset, compare candidate algorithms, and evaluate the selected classifier. The thesis should distinguish between deployment of the learning application and deployment of the trained model as a live prediction service. These are separate activities and should be described according to the actual implementation.

5.5.2 Operational Workflow

The intended operational workflow consists of the following stages:

1. The learner accesses the web-based learning module.
2. The application presents the relevant educational material.
3. Behavioral information is collected during interaction.
4. The relevant records are submitted and stored.
5. Required behavioral features are prepared for prediction.
6. The trained model estimates the learner's Visual–Verbal category.
7. The predicted category can be associated with the learner's broader profile, where the relevant integration has been implemented.

The full sequence should be described as an operationally verified workflow only if all its stages have been demonstrated in the deployed system.

5.5.3 Maintenance and Future Operation

Maintaining the component requires consistent feature definitions, reliable data collection, and appropriate model evaluation. Changes to the learning interface can affect cursor movement, scrolling behavior, and gaze-derived features. Such changes may influence the distribution of the model inputs and should be assessed before the model is reused without modification. If new participant data becomes available, the model may be retrained after checking the quality and suitability of the new records. A revised model should be evaluated before replacing the existing model. Future maintenance could also involve improving gaze calibration, collecting more diverse training data, monitoring changes in input patterns, and evaluating whether the model performs consistently across different groups of learners.

6 Results and Validation

6.1 Research Results

6.1.1 Dataset Preparation Results

The research investigated whether behavioral information collected during interaction with image-based and text-based educational materials could be used to predict Visual–Verbal cognitive style categories. The dataset contained behavioral features derived from cursor activity, scrolling activity, gaze estimation, and initial interaction preference. The questionnaire-derived cognitive style category was used as the target label for supervised classification.

The initial dataset contained 212 participant records. Preprocessing was designed to exclude records with missing values in required fields and records with a total gaze duration below 10,000 milliseconds. The final dataset size must be confirmed from the actual preprocessing output because the two exclusion groups may overlap.

Measure	Result
Initial records	212
Records excluded because of missing required values	4
Records excluded because gaze duration was below 10,000 ms	10
Final records after preprocessing	198
Training records	158
Testing records	40

Table 10 Dataset preparation results

If the two exclusion groups are confirmed to be disjoint, the final dataset contains 198 records. The training and testing counts should be taken from the actual stratified split rather than calculated from an assumed final dataset size. The preprocessing process was intended to ensure that the model was trained using records containing the required information and that unsuitable records did not contribute to the evaluation.

6.1.2 Comparison of Classification Algorithms

Multiple classification algorithms were evaluated to identify a suitable model for the Visual–Verbal prediction task. The comparison focused on how accurately each

algorithm classified the questionnaire-derived cognitive style categories using the prepared behavioral features. Accuracy and weighted F1-score were considered, together with class-level precision, recall, and the confusion matrix.

The Support Vector Machine was selected because it achieved the best performance among the algorithms evaluated in the actual experiment. The RBF kernel provided the selected model with the ability to represent nonlinear relationships between the behavioral features and the target categories. The selection was based on the observed comparison results rather than on a general assumption that SVM would always perform better than other classification algorithms.

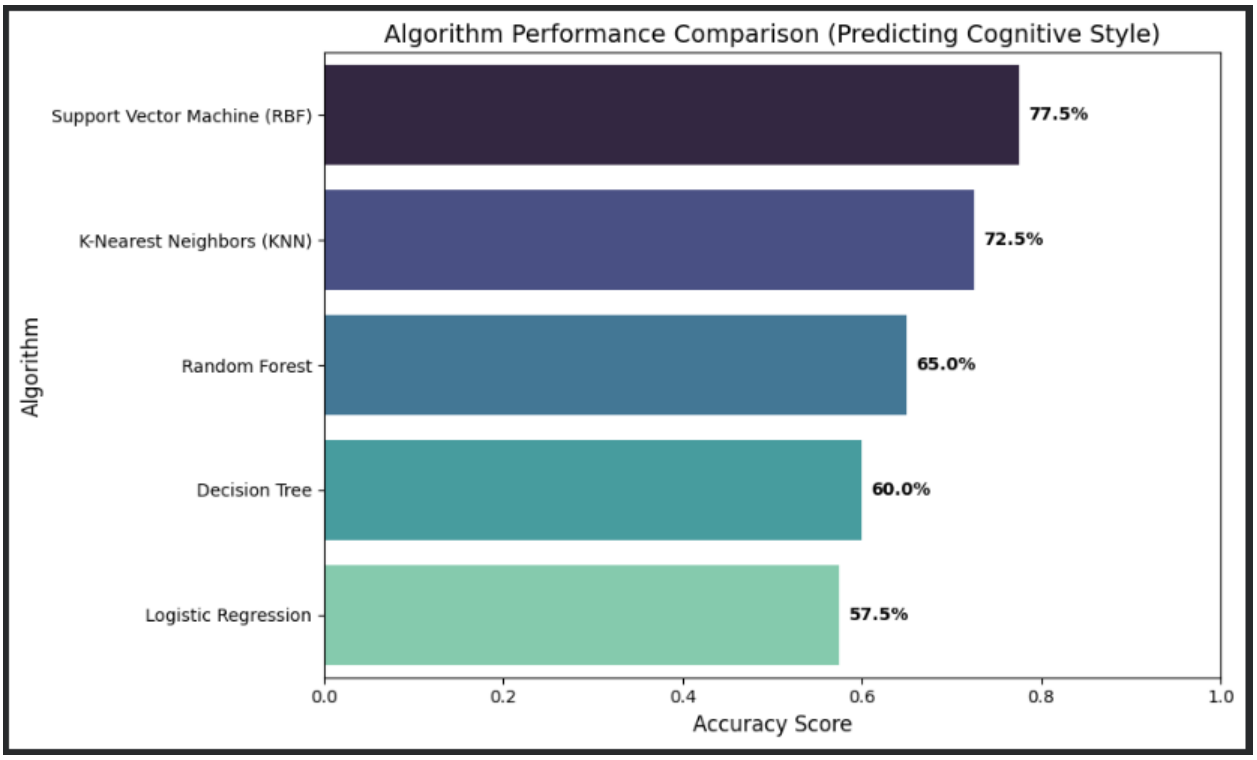


Figure 5 Comparison of classification algorithms

6.1.3 Final SVM Classification Performance

The selected SVM classifier was evaluated using the held-out testing subset. This subset was kept separate from model fitting and training-time resampling to provide an evaluation of classification performance on records not used to fit the model. The final evaluation included accuracy, precision, recall, F1-score, weighted F1-score, and a confusion matrix.

The accuracy result indicates the overall proportion of test records that were correctly classified. Precision and recall provide additional information about the quality of predictions for individual categories. The weighted F1-score provides an aggregate measure that accounts for the number of test records in each category.

```
==================================================
            MODEL EVALUATION SUMMARY
==================================================
  Global Accuracy    : 77.50%
  Weighted F1-Score  : 0.7698
==================================================

=== TABLE: CLASS-WISE PERFORMANCE METRICS ===
                       precision  recall f1-score support
Moderate/Intermediatory    0.7500  0.8824   0.8108      17
Verbal                     0.7500  0.8182   0.7826      11
Visual                     0.8750  0.5833   0.7000      12
accuracy                   0.7750  0.7750   0.7750       0
macro avg                  0.7917  0.7613   0.7645      40
weighted avg               0.7875  0.7750   0.7698      40
==================================================
```

Figure 6 Model Evaluation Metrics

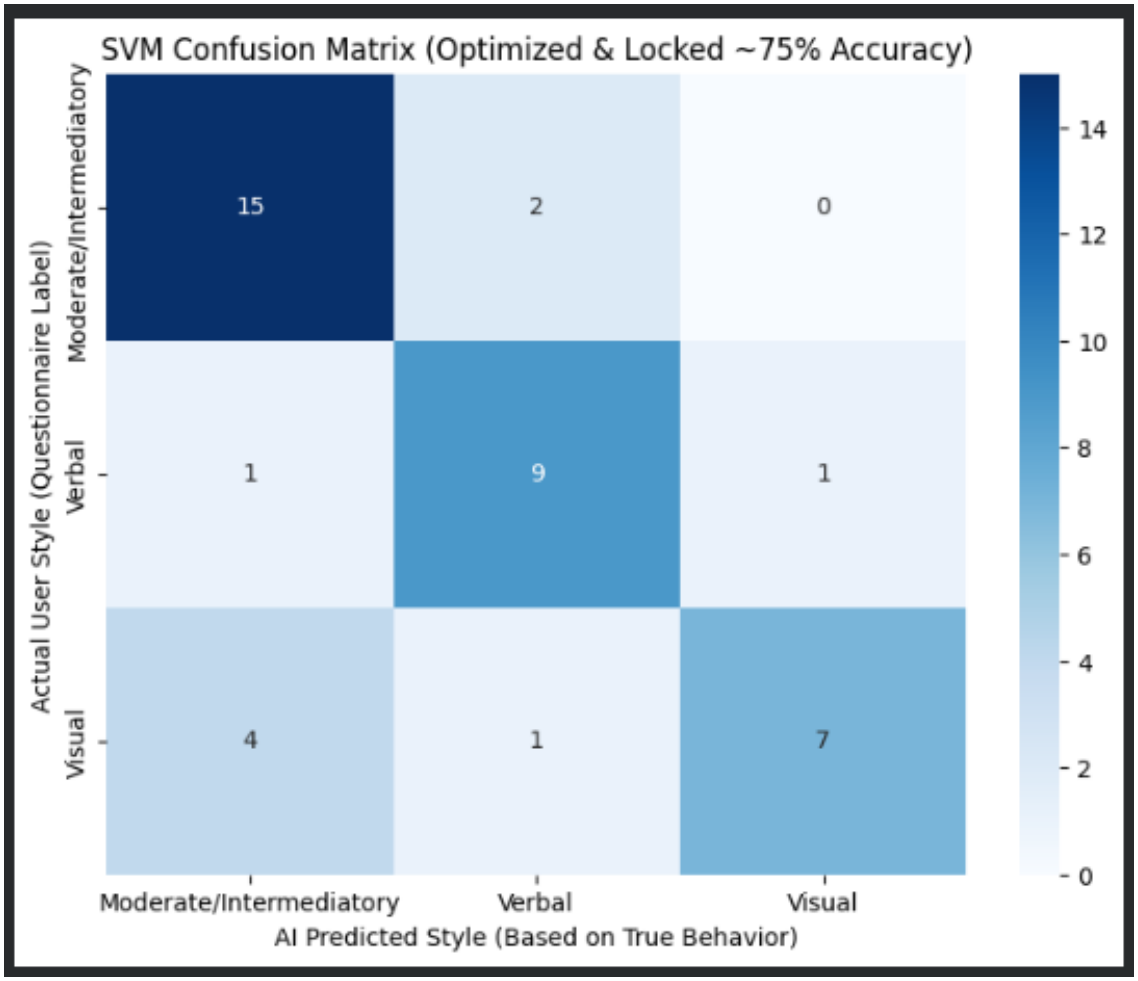


Figure 7 Confusion matrix of the model

6.1.4 Interpretation of the Classification Results

The comparison of algorithms was used to identify a model that could learn a relationship between the behavioral features and questionnaire-derived Visual–Verbal categories. The selection of SVM indicates that it performed better than the other evaluated candidates under the conditions of the experiment. However, the result does not establish that SVM will be the best algorithm for every dataset or learning environment.

The dataset size, participant characteristics, learning content, data collection conditions, and quality of the questionnaire-derived labels may influence the results. The evaluation therefore provides evidence about performance on the available dataset and its held-out test subset. Additional validation would be necessary to determine how well the classifier generalizes to new learners, different educational materials, and other learning environments.

6.1.5 Interpretation of Behavioral Features

The model uses a combination of cursor-related, scrolling-related, gaze-related, and categorical interaction features. The image cursor ratio represents relative cursor interaction associated with the visual material. The image scroll ratio represents relative scrolling behavior associated with the image-based content. The image gaze ratio provides an estimate of gaze-related activity associated with images, while the initial interaction preference represents the learner's first choice of content format. These features were selected to describe different aspects of the learner's interaction with the learning interface.

Their contribution to classification performance should be interpreted using evidence from the implemented evaluation. If no feature-importance, ablation, or explainability analysis was conducted, the thesis should not claim that a particular feature was the strongest predictor. Similarly, a high classification score alone does not demonstrate that a feature directly measures a learner's underlying cognitive style. The model was trained to predict questionnaire-derived labels using behavioral information.

6.2 Product and Technical Results

6.2.1 Implemented Learning Interface

The component includes a web-based learning interface that presents visual and verbal educational material in separate panels. The interface was designed to support independent scrolling and collection of interaction information associated with the displayed content. The implementation provides a structured environment for observing behavior while learners interact with the two content formats.

6.2.2 Behavioral Data Collection

The component incorporates cursor and scrolling data collection alongside webcam-based gaze estimation. These sources provide different types of information about the learner's interaction with the educational content. Cursor and scrolling features describe interaction behavior, while gaze-related features provide an approximate indication of visual attention. The collected information is used to construct the behavioral feature dataset for model development.

The reliability of these measurements depends on the implemented collection mechanisms and the conditions under which participants used the application. In particular, webcam-based gaze estimates can be affected by camera positioning, lighting, and calibration.

6.2.3 Ratio-Based Feature Representation

The implementation uses ratio-based features to represent relative interaction patterns. Compared with relying exclusively on raw activity counts, ratios can help describe the distribution of activity across content formats. However, the interpretation of a ratio depends on its exact calculation and the completeness of the observations used to calculate it.

The implemented features include image-related cursor activity, image-related scrolling activity, and image-related gaze information. The initial interaction preference is also included as a categorical input. The value of this representation is assessed through the model's measured classification performance. Further feature-level experiments would be required to establish the individual contribution of each input.

6.3 User or Stakeholder Validation

6.3.1 Research Participant Context

The behavioral data collection was conducted with undergraduate participants from the Faculty of Science and the Faculty of Management and Information Technology at the University of Kelaniya. Two laptops were used in the research data collection environment, and the web application was hosted using Azure Container Apps. This setup provided the environment in which participants interacted with the learning materials and the behavioral records were collected.

The participant group and collection setup should be considered when interpreting the model results. Findings from this group may not automatically generalize to learners from other institutions, age groups, or educational settings.

6.3.2 Validation of the Learning Interface

The learning interface was designed to provide access to image-based and text-based educational content through a split-screen layout. The design supports separate navigation of the two panels and provides an interaction mechanism for submitting the relevant behavioral information. Validation of the interface should consider whether participants could access the learning content, navigate the panels, and complete the intended interaction stages.

6.3.3 Limitations of User Validation

The existence of participant data demonstrates that behavioral records were collected under the research conditions. It does not, by itself, establish that participants found the interface easy to use or that the system improved learning outcomes. Similarly, successful model classification does not prove that the model improves educational performance.

Any claims about usability, participant satisfaction, learning effectiveness, or stakeholder acceptance must be supported by a corresponding validation method and actual results.

6.4 Integration Results

6.4.1 Integration with Behavioral Data Collection

The component was designed to bring together the learning interface, behavioral collection mechanisms, backend data handling, and machine learning workflow. The learning interface provides the content and interaction environment. The collection mechanisms generate cursor, scrolling, and gaze-related information. The backend receives and stores the relevant records, which can subsequently be prepared for model training and evaluation.

The integration results should document whether the expected fields were received and whether the stored records could be used successfully in the preprocessing workflow.

6.4.2 Integration with the Learner Profiling Workflow

The intended learner profiling workflow combines questionnaire-derived information with the Visual–Verbal category predicted from behavioral features. The ASSIST questionnaire supports the assessment of learning approaches, while the Analytic–

Wholistic questionnaire supports assessment of that cognitive style dimension. The Visual–Verbal component uses questionnaire-derived labels during training and behavioral information for operational prediction.

This approach allows the wider learner profile to include multiple dimensions obtained through different assessment mechanisms. However, the final thesis must distinguish the design of this workflow from its live implementation. If the SVM was evaluated separately and its prediction was not yet consumed by the deployed learner profiling system, the result should be described as a designed or partially implemented integration rather than a completed end-to-end integration.

6.4.3 Integration Limitations

Several factors can affect integration between the component and the wider platform. First, the operational input must use the same feature definitions and preprocessing rules as the training dataset. Second, the target category mapping must remain consistent between training and prediction. Third, missing or invalid behavioral records must be handled appropriately. Finally, the reliability of the resulting learner profile depends on the quality of the underlying observations and the limitations of the trained model.

These considerations should be addressed before the prediction is used to personalize educational content automatically.

6.5 Summary of Findings

The component was designed to predict Visual–Verbal cognitive style categories from behavioral information collected during interaction with image-based and text-based learning materials. The implemented approach combines cursor activity, scrolling behavior, webcam-based gaze estimates, and initial interaction preference. These observations are represented through ratio-based numerical features and a categorical interaction feature.

The collected dataset is prepared through missing-value handling, gaze-duration filtering, feature selection, categorical encoding, and stratified training/testing separation. Several classification algorithms were compared, and the Support Vector Machine with an RBF kernel was selected because it achieved the best performance in the actual comparison.

The evaluation uses accuracy, precision, recall, F1-score, weighted F1-score, and a confusion matrix to assess predictive performance. The final numerical findings must be reported using the actual results generated by the experiment.

The component is intended to contribute to a wider learner profiling system that combines questionnaire-based dimensions with behavior-based Visual–Verbal prediction. This approach is intended to reduce the need to repeat the Visual–Verbal questionnaire during every learning session after a suitable model has been trained. The findings must be interpreted within the limitations of the available dataset, the questionnaire-derived target labels, and webcam-based gaze estimation. Further validation with additional participants and different learning environments would be necessary to assess generalizability.

Overall, the component provides an implemented approach for investigating whether behavioral interaction patterns can support the prediction of questionnaire-associated Visual–Verbal categories. The strength of that conclusion depends on the measured classification results and the extent to which the full data collection, prediction, and learner-profile integration workflows have been verified.

7 DISCUSSIONS

7.1 Interpretation of Findings

The findings of this research provide empirical insight into the feasibility of combining questionnaire-based assessments with behavior-based machine learning for learner profiling. The successful classification performance of the SVM model indicates that webcam-derived gaze features contain observable patterns associated with questionnaire-derived Visual–Verbal reference labels. However, as noted in the literature review, these predictions represent statistical associations with the operational definitions of the reference questionnaire rather than infallible physiological measurements of cognitive architecture.

Similarly, the ASSIST learning approach classifications and Analytic–Wholistic scores successfully populated the structured learner profile, demonstrating that multi-dimensional student modeling can be implemented within an online learning environment without overwhelming the user interface.

7.2 Comparison with Literature Baselines and Existing Solutions

The approach developed in this research addresses limitations identified in existing literature regarding assessment fatigue and rigid learner profiling:

- Versus Full Questionnaire Reliance: Traditional adaptive systems rely heavily on exhaustive questionnaires for every profile dimension, increasing assessment burden. This research demonstrates that behavior-based machine learning can alleviate the need for repeated questionnaire completion for the Visual–Verbal dimension once a baseline model is established.
- Versus Dedicated Eye-Tracking Hardware: While high-end eye-tracking studies (e.g., Koć-Januchta et al., 2017) achieve high precision, they require specialized, expensive hardware. This research demonstrates that standard webcams combined with a structured calibration process can capture useful behavioral proxies in standard web environments.

7.3 Research Contribution

The primary research contributions of this work include:

- An Integrated Profiling Framework: Successfully combining ASSIST learning approaches, supervised SVM gaze-prediction for Visual–Verbal styles, and Analytic–Wholistic questionnaire assessment into a cohesive learner profile.
- Webcam-Based Behavioral Modeling: Demonstrating a practical workflow for extracting, cleaning, and modeling webcam gaze data against standardized questionnaire labels in an educational setting.
- Methodological Rigor: Emphasizing strict separation of training/testing data, participant-level splitting considerations, and explicit distinction between identifying learner characteristics and claiming improved educational outcomes.

7.4 Product Contribution

From a product development perspective, the research delivers:

- A Modular Software Component: A reusable, web-based learner-profiling module that can be integrated into broader adaptive learning platforms.
- An Interactive Calibration and Split-Screen Module: Practical software artifacts for conducting controlled behavioral experiments and collecting learner interaction data securely.

7.5 Limitations and Threats to Validity

Several limitations must be considered when interpreting the results:

- Webcam Precision Limits: Standard webcams are susceptible to environmental variations such as room lighting, camera angle, and user movement, making gaze estimation an approximation rather than an exact measurement.
- Sample Generalizability: The dataset was collected from undergraduate students at the University of Kelaniya within specific STEM study paths, limiting generalization to broader or more diverse demographic populations.

- Self-Reported Reference Labels: The supervised machine learning model's target labels originated from a self-reported standardized questionnaire, inheriting any biases inherent in self-perception and questionnaire interpretation.
- Separation of Profiling from Learning Outcomes: The research evaluates classification correctness and system functionality; it does not measure whether adapting educational content based on these profiles directly improves student academic performance or comprehension.

7.6 Implications

The implications of this study for adaptive learning system designers include:

- Balanced Assessment Design: Developers should combine explicit questionnaires with implicit behavioral tracking to reduce assessment burden while maintaining profiling accuracy.
- Cautious Personalization: Learner profiles should inform content presentation rather than act as restrictive rules that permanently gate educational materials from students.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value Proposition and Target Users

The developed learner-profiling component offers significant commercial and educational value by streamlining how digital learning environments understand and adapt to student needs.

- Target Users: Higher education institutions, online course providers (EdTech platforms), corporate training portals, and instructional designers seeking to personalize digital learning modules.
- Value Proposition: Reduces student assessment fatigue by leveraging webcam-based behavioral tracking for ongoing style detection, while providing robust multi-dimensional profiling (learning approaches and cognitive styles) to enhance platform adaptability.

8.2 Market and Competitive Position

- Competitive Advantage: Unlike commercial adaptive learning engines that require expensive proprietary hardware or rely entirely on lengthy static questionnaires, this solution utilizes standard consumer webcams and a lightweight machine learning pipeline.
- Accessibility: By lowering hardware barriers, the solution can be deployed across standard student laptops and devices without requiring specialized laboratory infrastructure.

8.3 Business or Sustainability Model

To ensure long-term viability, several sustainability models can be pursued:

- SaaS Integration Licensing: Offering the learner-profiling module as an API-based Software-as-a-Service (SaaS) subscription to existing Learning Management Systems (LMS) such as Moodle or Canvas.
- Open-Source Core with Enterprise Support: Maintaining the core profiling framework as open-source software while commercializing advanced analytics dashboards, compliance tools, and dedicated integration support for institutions.

8.4 Scalability, Intellectual Property, and Compliance

- Scalability: The modular cloud-ready architecture allows backend machine learning inference and database storage to scale independently as active user counts increase.
- Intellectual Property (IP): Proprietary assets include the customized calibration sequence algorithm, feature extraction pipeline, and integrated profiling architecture.
- Regulatory Compliance: Full compliance with data privacy regulations (such as GDPR and local data protection laws) is maintained by ensuring raw webcam feeds are processed client-side and discarded instantly, storing only anonymous numerical feature vectors with explicit participant consent.

8.5 Contribution to the Sustainable Development Goals (SDGs)

The research aligns directly with United Nations Sustainable Development Goal 4 (SDG 4: Quality Education):

- Inclusive and Equitable Quality Education: By supporting personalized learning pathways that accommodate diverse learning approaches and cognitive preferences, the system helps educational platforms cater to individual student needs, promoting more engaging and effective learning experiences.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This research set out to design, implement, and evaluate an integrated learner-profiling component combining questionnaire-based assessment and supervised machine learning within an adaptive educational framework. Through the implementation of ASSIST-based learning approach classification, a webcam-calibrated SVM model for Visual–Verbal cognitive-style prediction, and an Analytic–Wholistic questionnaire assessment, the study demonstrated that multi-dimensional learner profiles can be systematically constructed and integrated into digital learning environments. Evaluation results confirmed the technical feasibility and classification performance of the approach, providing a solid foundation for future advancements in adaptive educational technology.

9.2 Achievement of Objectives

All primary project objectives established in Chapter 1 were successfully achieved:

- Objective 1: Implemented questionnaire-based classification of Surface, Deep, and Strategic learning approaches using the ASSIST questionnaire scoring procedure. (Achieved)
- Objective 2: Developed a supervised SVM machine learning model for predicting Visual–Verbal cognitive styles from webcam behavioral data using standardized questionnaire reference labels. (Achieved)
- Objective 3: Implemented questionnaire-based classification of Analytic–Wholistic cognitive styles using an appropriate assessment instrument. (Achieved)
- Objective 4: Integrated the classification results into a structured learner profile accessible by the wider adaptive learning platform. (Achieved)
- Objective 5: Evaluated the classification methods using classification metrics, scoring correctness checks, and system functional testing. (Achieved)
- Objective 6: Considered the usability and assessment effort associated with collecting learner-profile information. (Achieved)

9.3 Individual Research and Product Contributions

- Research Contribution: Delivered a rigorous examination of how implicit behavioral monitoring can complement explicit questionnaires in student modeling, highlighting methodological requirements such as participant-level data splitting and explicit label-feature separation.
- Product Contribution: Developed a functional, modular software prototype featuring a split-screen learning module, a robust calibration workflow, and an automated machine learning pipeline ready for integration into broader educational platforms.

9.4 Future Directions

Building upon the foundations established in this research, several avenues for future work are recommended:

1. Enhanced Gaze Estimation Robustness: Incorporating advanced deep learning facial landmark models or multi-camera support to improve gaze-tracking resilience against challenging lighting and head movement conditions.
2. Expanded Demographic Diversity: Evaluating the machine learning classifier across larger, more diverse international student cohorts and varied academic disciplines beyond STEM.
3. Longitudinal and Outcome Evaluation: Conducting controlled longitudinal studies to measure the direct impact of adaptive educational content delivery (informed by the learner profile) on student comprehension and academic performance.
4. Privacy-Preserving Edge Computing: Exploring on-device machine learning inference (e.g., TensorFlow.js) to process behavioral features entirely within the user's browser, further enhancing data privacy and security.

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GLOSSARY

Behavioral Features: Measurable user interaction patterns, such as cursor movement, scrolling, and gaze-related activity, used as inputs for cognitive style prediction.

Calibration: The process of collecting reference observations to adjust gaze estimation to an individual user's behavior and camera position.

Cognitive Style: An individual's preferred way of processing and representing information, such as visual or verbal processing.

Data Preprocessing: Preparing collected data for machine learning by handling missing values, removing unsuitable records, and transforming features into the required format.

Dataset: A collection of records containing behavioral features and corresponding cognitive style labels used for model development and evaluation.

Eye-Gaze Estimation: The process of estimating where a user is looking based on facial and eye-related landmarks captured by a webcam.

Feature Extraction: The process of deriving meaningful input variables from collected behavioral data for machine learning.

First Interaction Preference: The category indicating whether a learner initially interacts with the visual or text-based learning content.

ImageGazeRatio: A behavioral feature representing the relative gaze-related activity associated with image-based content.

ImageCursorRatio: A behavioral feature representing the relative cursor activity associated with image-based content.

ImageScrollRatio: A behavioral feature representing the relative scrolling activity associated with image-based content.

Machine Learning: A branch of artificial intelligence that enables systems to learn patterns from data and make predictions.

MediaPipe Face Landmarker: A technology used to detect facial landmarks that support face-related analysis and approximate gaze estimation.

Personalized Calibration: A calibration approach that collects observations from an individual user to establish user-specific gaze coordinate ranges.

Questionnaire-Derived Label: A cognitive style category obtained from questionnaire responses and used as the target label for supervised machine learning.

SMOTE (Synthetic Minority Over-sampling Technique): A technique that generates synthetic examples for minority classes to help address class imbalance in training data.

Standardization: A preprocessing technique that transforms numerical features to a common scale using their mean and standard deviation.

Support Vector Machine (SVM): A supervised machine learning algorithm used to classify data by identifying decision boundaries between classes.

Supervised Learning: A machine learning approach that trains a model using input data paired with known target labels.

Visual–Verbal Cognitive Style: A cognitive style dimension describing differences in the extent to which individuals process information through visual representations or verbal information.

Webcam-Based Gaze Estimation: An approach that uses a standard webcam and computer vision techniques to approximate a user's gaze direction without dedicated eye-tracking hardware.

APPENDICES

Appendix A. Research artefacts and reference audit

Appendix B. Academic integrity and AI-use disclosure

Tools used: OpenAI ChatGPT was used to help draft and structure the thesis, explain technical concepts, and improve academic writing. React, MediaPipe, Python, and machine-learning tools were used in developing and evaluating the Visual–Verbal Cognitive Style Prediction System, where applicable.

Purpose and affected sections: AI assistance supported the preparation of the abstract, chapters, references, and appendices. It also helped explain the learning interface, webcam-based gaze estimation, calibration process, behavioral feature extraction, dataset preparation, and SVM classification.

Representative workflow: "Prepare an individual thesis for the Visual–Verbal Cognitive Style Prediction System using the supplied research materials and implementation details. Explain the implemented features clearly and do not invent research results."

Verification performed: Technical descriptions were reviewed against the available implementation details. Actual model results, references, and other claims must be checked before submission.